DELPHIX CONTINUOUS COMPLIANCE · DISCOVERY POLICY

Peru — Ley 29733 de Protección de Datos Personales

Discovery and masking of Peruvian personal data: DNI and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 2, 5 — including economic income —, health data, financial data and criminal records.

85	146	119
DATA CLASSES	CLASSIFIERS	ALGORITHMS

Discovery policy `PE – Ley 29733 – v3`

Version 3 of October 12, 2026 · Assignment threshold 60%

Contents

The first sections explain what the policy is for and how it works. The data class chapters then describe every data class: what it is, how its data is found and how it is masked, with examples. The appendices list every algorithm, file and classifier the policy contains.

1	**About this discovery policy**	
2	**Ley 29733 in brief**	
3	**Dissociation, anonymization and non-production copies**	
4	**How sensitive data is discovered**	
5	**How data is masked**	
6	**Essential and extended packs**	
7	**Data classes at a glance**	

PART 8 · DIRECT IDENTIFIERS

8.1	**DNI**	PE_L1_DNI
8.2	**RUC**	PE_L1_RUC
8.3	**Verification character**	PE_L1_DIGITO_VERIFICACION
8.4	**Foreigner documents**	PE_L1_CARNE_EXTRANJERIA
8.5	**Passport**	PE_L1_PASAPORTE
8.6	**Civil registry records**	PE_L1_ACTA
8.7	**EsSalud, AFP and ONP**	PE_L1_SEGURO_SOCIAL
8.8	**Student code**	PE_L1_CODIGO_ESTUDIANTE
8.9	**Professional registration**	PE_L1_COLEGIATURA
8.10	**Given name**	PE_L1_NOMBRE
8.11	**Surname**	PE_L1_APELLIDO
8.12	**Full name**	PE_L1_NOMBRE_COMPLETO
8.13	**Business name**	PE_L1_RAZON_SOCIAL
8.14	**E-mail**	PE_L1_EMAIL
8.15	**Phone**	PE_L1_TELEFONO
8.16	**Address**	PE_L1_DIRECCION
8.17	**Address complement**	PE_L1_DIRECCION_COMPLEMENTO
8.18	**Bank account**	PE_L1_CUENTA_BANCARIA
8.19	**Payment card**	PE_L1_TARJETA
8.20	**Digital wallet**	PE_L1_BILLETERA
8.21	**IP address**	PE_L1_IP
8.22	**Device identifier**	PE_L1_DISPOSITIVO
8.23	**Cookies and tracking identifiers**	PE_L1_COOKIE
8.24	**Vehicle plate**	PE_L1_PLACA

8.25	Vehicle and licence	PE_L1_VEHICULO
8.26	Property	PE_L1_INMUEBLE
8.27	Customer and contract numbers	PE_L1_CONTRATO
8.28	User and social media	PE_L1_USUARIO
8.29	Credentials	PE_L1_CREDENCIAL
8.30	Geolocation	PE_L1_GEOLOCALIZACION
8.31	Free text	PE_L1_TEXTO_LIBRE

PART 9 · QUASI-IDENTIFIERS

9.1	Birth date	PE_L2_FECHA_NACIMIENTO
9.2	Birth year	PE_L2_ANIO_NACIMIENTO
9.3	Age	PE_L2_EDAD
9.4	Personal event dates	PE_L2_FECHA_EVENTO
9.5	Sex	PE_L2_SEXO
9.6	Distrito and provincia	PE_L2_DISTRITO
9.7	UBIGEO	PE_L2_UBIGEO
9.8	Centro poblado and caserío	PE_L2_CENTRO_POBLADO
9.9	Urbanización and asentamiento humano	PE_L2_URBANIZACION
9.10	Postal code	PE_L2_CODIGO_POSTAL
9.11	Marital status	PE_L2_ESTADO_CIVIL
9.12	Occupation	PE_L2_OCUPACION
9.13	Employer	PE_L2_EMPLEADOR
9.14	Education	PE_L2_NIVEL_EDUCATIVO
9.15	School	PE_L2_INSTITUCION_EDUCATIVA
9.16	Dependants and children	PE_L2_PERSONAS_A_CARGO

PART 10 · SENSITIVE DATA (ART. 2, 5)

10.1	Ethnic origin	PE_L3_ORIGEN_ETNICO
10.2	Indigenous people	PE_L3_PUEBLO_INDIGENA
10.3	Language	PE_L3_LENGUA
10.4	Nationality	PE_L3_NACIONALIDAD
10.5	Religion	PE_L3_RELIGION
10.6	Philosophical and moral convictions	PE_L3_CONVICCION_FILOSOFICA
10.7	Political opinion	PE_L3_OPINION_POLITICA
10.8	Party membership	PE_L3_AFILIACION_PARTIDARIA
10.9	Trade union membership	PE_L3_AFILIACION_SINDICAL
10.10	Biometric data	PE_L3_BIOMETRICO
10.11	Genetic data	PE_L3_GENETICO

10.12	Sexual orientation	PE_L3_ORIENTACION_SEXUAL
10.13	Sexual life	PE_L3_VIDA_SEXUAL
10.14	Gender identity	PE_L3_IDENTIDAD_GENERO
10.15	Victims	PE_L3_VICTIMA

PART 11 · HEALTH DATA

11.1	Health identifiers	PE_SALUD_IDENTIFICADOR
11.2	Health insurance	PE_SALUD_COBERTURA
11.3	Diagnosis and ICD-10	PE_SALUD_DIAGNOSTICO
11.4	Procedures	PE_SALUD_PROCEDIMIENTO
11.5	Medicines	PE_SALUD_MEDICAMENTO
11.6	Clinical text	PE_SALUD_TEXTO_CLINICO
11.7	Test results	PE_SALUD_RESULTADO_EXAMEN
11.8	HIV and sexually transmitted infections	PE_SALUD_VIH
11.9	Blood group and Rh	PE_SALUD_GRUPO_SANGUINEO
11.10	Disability	PE_SALUD_DISCAPACIDAD
11.11	Occupational health	PE_SALUD_OCUPACIONAL
11.12	Sexual and reproductive health	PE_SALUD_REPRODUCTIVA
11.13	Mental health	PE_SALUD_MENTAL

PART 12 · FINANCIAL AND SOCIOECONOMIC DATA

12.1	Income and pay	PE_FIN_INGRESOS
12.2	Credit history	PE_FIN_HISTORIAL_CREDITICIO
12.3	Debt	PE_FIN_DEUDA
12.4	Assets	PE_FIN_PATRIMONIO
12.5	Pension and benefits	PE_FIN_PENSION
12.6	Social programmes	PE_FIN_PROGRAMA_SOCIAL
12.7	SISFOH classification	PE_FIN_CLASIFICACION_SOCIOECONOMICA
12.8	Housing tenure	PE_FIN_VIVIENDA

PART 13 · CRIMINAL RECORDS

13.1	Criminal records	PE_PENAL_ANTECEDENTES
13.2	Court or prosecution file	PE_PENAL_EXPEDIENTE
14	The DNI and RUC check digits	
15	Where people live: distritos, UBIGEO and postal code	
16	Limits	
17	Verification	
18	Sources	

A	Algorithms
B	Files
C	Exclusion lists
D	Discovery policy and classifiers

1

About this discovery policy

This discovery policy finds the personal data in a Peruvian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and DNIs alone is not enough: a birth date, a sex and a small distrito already point at one person. The policy works in the layers where the risk is:

- **Direct identifiers** — DNI and RUC, foreigner card, passport, civil registry records, EsSalud and pension numbers, student code, professional registration, names, contact, address, accounts, cards, wallets, devices, cookies, plates, property, free text.
- **Quasi-identifiers** — birth date, age, sex, distrito and provincia with their UBIGEO, postal code, urbanización, centro poblado, marital status, occupation, employer, education.
- **Sensitive data** — those of article 2, numeral 5: biometric data that identifies on its own, racial and ethnic origin, **economic income**, political, religious, philosophical and moral opinions or convictions, trade union membership, and information on health or sexual life; plus genetic data, gender identity, indigenous people and language, nationality and victim status.
- **Health data** — sensitive without qualification, in a group of its own: clinical record, insurance, ICD-10, procedures, medicines, HIV, blood group, disability, occupational, reproductive and mental health.
- **Financial data** — the income the law itself calls sensitive, plus credit history, debt, assets, pensions, social programmes, the SISFOH classification and housing tenure.
- **Criminal records** — record certificates, procedural situation and court or prosecution file numbers.

In Data Control Tower the policy is one discovery policy with its classifiers, the data classes they vote for, the algorithms those data classes name — and the algorithms those rely on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — `NOMBRE`, `DOCUMENTO`, `EXPEDIENTE`, `CARGO` — needs the values to agree. A column with a meaningless name (`CAMPO1`, `OBS`) is still found when its values have a recognizable shape or vocabulary: a DNI, an eleven-digit RUC, a nine-digit mobile, an address with urbanización, manzana and lote, a plate, a distrito.

Column names are recognized in Spanish, in the abbreviations of Peruvian systems (`NUM_DOC`, `FEC_NAC`, `UBIGEO`, `RUC`, `ESSALUD`, `CODIGO_MODULAR`) and in English.

> Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding processes often keep a photo of the DNI — with the fingerprint and the signature — and a validation selfie. Report those columns as residual risk.

2

Ley 29733 in brief

The **Ley 29733 de Protección de Datos Personales** was published on 3 July 2011 and developed by a regulation of 2013. On **30 November 2024** a new regulation replaced it, **Decreto Supremo 016-2024-JUS**, in force since **30 March 2025**. Applying and supervising both falls to the **Autoridad Nacional de Protección de Datos Personales (ANPD)**, within the Ministry of Justice and Human Rights.

What matters for a copy of production data:

- **Personal data** is any information about a natural person that identifies them or makes them identifiable through means that can reasonably be used (art. 2, 4). The data subject is always a natural person (art. 2,

12).

- **Sensitive data** is a short and blunt list (art. 2, 5): biometric data that by itself can identify the subject; data on racial and ethnic origin; **economic income**; political, religious, philosophical or moral opinions or convictions; trade union membership; and information on health or sexual life. Income appears in no neighbouring law: in Peru a salary column carries the weight of a diagnosis.
- **Anonymization** makes the subject unidentifiable in a way that **cannot be reversed** (art. 2, 14). **Dissociation** is the procedure that prevents identification, and article 41.1 of the 2024 regulation states outright that **this procedure is reversible**. Article 41.2 asks the controller to choose the technique that fits the data, the number of subjects and the risk.
- **Where they appear.** Using data for a purpose other than the one collected is a serious infringement — *unless an anonymization or dissociation procedure has taken place* (art. 133.7 of the regulation). That sentence is the legal basis for a masked non-production copy.
- **Security.** The controller keeps security measures in place (arts. 39 and 40 of the regulation ask for an impact assessment where the risk is high), appoints a **Personal Data Protection Officer** where articles 37 to 39 require one, and notifies a breach to the ANPD within **48 hours** of becoming aware of it (art. 34).
- **Minors.** Data of a child under fourteen needs the consent of the parents; between fourteen and eighteen the adolescent consents themselves (art. 25 of the regulation). A copy that keeps exact birth dates keeps that distinction visible.
- **Sanctions.** Fines are measured in UIT and reach the tens of unidades impositivas tributarias for very serious infringements, with the ANPD able to order the processing to stop.

What the law leaves out. The law does not apply to data held by natural persons for domestic purposes, nor to data processed for national defence, public security or the investigation of crimes, nor — as the regulation makes explicit — to anonymized data.

3

Dissociation, anonymization and non-production copies

Peru is one of the few countries that writes the difference down in the law itself. **Anonymization** is irreversible (art. 2, 14). **Dissociation** is the procedure that prevents identification, and article 41.1 of the 2024 regulation says in so many words that it **is reversible**.

The algorithms of this policy are deterministic for the Delphix masking key: the same DNI masks the same way in every table, which keeps joins working. That key is what makes the procedure reversible. The masked copy is **dissociated** in the sense of article 41, not **anonymized**, and remains personal data for whoever holds the key and production.

- Keep the engine key apart from the copy, and treat whoever holds both as processing personal data.
- Article 133.7 of the regulation excuses the change of purpose when a dissociation procedure has taken place; article 41.2 asks for the technique to fit the data, the number of subjects and the risk. Record which technique this policy applies to each column — the appendices list them one by one.
- If what you need is anonymized data — irreversible, out of the law altogether —, measure the re-identification risk of the quasi-identifiers that remain; article 2, 14 is a higher standard than this policy meets.

4

How sensitive data is discovered

A discovery job looks at every column of its rule set and asks each classifier of the discovery policy how confident it is that the column belongs to its data class. The policy uses the four classifier frameworks of Delphix:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the data class out. It never assigns a data class on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per data class, and the column is assigned to the data class with the highest confidence when it reaches the assignment threshold of the discovery policy — **60%** here. Values that do not match a format or a list count against their data class by the classifier's reject strength.

These are the names Data Control Tower gives the objects: data classes, classifiers and a discovery policy, run by a discovery job. A Continuous Compliance Engine used on its own calls the same objects domains, classifiers and a profile set, run by a profiling job; everything in this document applies to both.

Databases pad values: a CHAR column fills them with spaces up to its length, and a numeric code kept as text often comes with zeros on the left. Discovery tests values as they come, so every format pattern first removes the spaces around the value, patterns of digits also accept zeros on the left, lists of single words are looked up word by word, and lists of names made of several words (places, categories) are held as a single value-format pattern. A column whose name says nothing — `CAMPO1`, `FLD_07`, the short names of packaged systems — is found by its values alone wherever their format is distinctive: documents with a check digit, e-mail addresses, telephone numbers, cards, places, names.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters: `rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

> Discovery proposes; it does not decide. Review the data classes assigned in the rule set before running a masking job, and look for columns the policy left unassigned.

5

How data is masked

Each data class names one algorithm, which a masking job applies to every column the rule set assigns to the data class. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each data class shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine and compliance node uses the same keys.

Where an algorithm that comes with Delphix already does the job — CM Numeric for amounts, Credit Card, the blood group lookup — the policy names it instead of creating one of its own. Those algorithms appear in the trees by their `dlpx-core:` name, the one an engine uses; Data Control Tower lists them without the

prefix (`CM Numeric` , `CreditCard` , `BloodType SL`). They are on every engine and in every Data Control Tower, and are not listed in Appendix A.

Padding stays where it is: the spaces around a value are kept, and zeros on the left of a number are not masked — `0012345678` becomes `00` followed by what `12345678` becomes. A value kept in a zero-filled CHAR column still joins the same value kept as a number.

> The masked examples in this document were produced with one particular key. With other keys the masked values differ; their shape does not.

6

Essential and extended packs

The policy can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create just its data classes, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the discovery policy.

PACK	DATA CLASSES	CLASSIFIERS	ALGORITHMS	FILES
Essential	12	25	30	5
Extended	85	146	119	58

The essential pack holds these data classes, marked **Essential** in the tables and chapters:

- DNI — `PE_L1_DNI`
- RUC — `PE_L1_RUC`
- Foreigner documents — `PE_L1_CARNE_EXTRANJERIA`
- Passport — `PE_L1_PASAPORTE`
- Given name — `PE_L1_NOMBRE`
- Surname — `PE_L1_APELLIDO`
- Full name — `PE_L1_NOMBRE_COMPLETO`
- E-mail — `PE_L1_EMAIL`
- Phone — `PE_L1_TELEFONO`
- Address — `PE_L1_DIRECCION`
- Address complement — `PE_L1_DIRECCION_COMPLEMENTO`
- Birth date — `PE_L2_FECHA_NACIMIENTO`

> With fewer data classes competing, a column that a data class of the extended pack would claim by its name can be proposed for an essential data class instead — a numeric code, for instance, can look like a telephone number. Review the discovery result.

7

Data classes at a glance

Every data class of the policy, how it is found and the algorithm that masks it.

DATA CLASS	FOUND BY	ALGORITHM
DIRECT IDENTIFIERS		
DNI ESSENTIAL PE_L1_DNI	COLUMN NAME · DATA TYPE · VALUE FORMAT	PE_DOCUMENTO
RUC ESSENTIAL PE_L1_RUC	COLUMN NAME · VALUE FORMAT	PE_RUC_COLUMNA
Verification character PE_L1_DIGITO_VERIFICACION	COLUMN NAME · DATA TYPE	PE_CM_DIGITOS_LETRAS
Foreigner documents ESSENTIAL PE_L1_CARNE_EXTRANJERIA	COLUMN NAME	PE_CM_ALFANUM
Passport ESSENTIAL PE_L1_PASAPORTE	COLUMN NAME · VALUE FORMAT	PE_CM_ALFANUM
Civil registry records PE_L1_ACTA	COLUMN NAME	PE_CM_ALFANUM
EsSalud, AFP and ONP PE_L1_SEGURO_SOCIAL	COLUMN NAME	PE_CM_ALFANUM
Student code PE_L1_CODIGO_ESTUDIANTE	COLUMN NAME	PE_CM_ALFANUM
Professional registration PE_L1_COLEGIATURA	COLUMN NAME	PE_CM_ALFANUM
Given name ESSENTIAL PE_L1_NOMBRE	COLUMN NAME · KNOWN VALUES	PE_NOMBRES
Surname ESSENTIAL PE_L1_APELLIDO	COLUMN NAME · KNOWN VALUES	PE_APELLIDOS
Full name ESSENTIAL PE_L1_NOMBRE_COMPLETO	COLUMN NAME · KNOWN VALUES	PE_NOMBRE_COMPLETO
Business name PE_L1_RAZON_SOCIAL	COLUMN NAME	PE_RAZON_SOCIAL
E-mail ESSENTIAL PE_L1_EMAIL	COLUMN NAME · DATA TYPE · VALUE FORMAT	PE_EMAIL
Phone ESSENTIAL PE_L1_TELEFONO	COLUMN NAME · VALUE FORMAT	PE_TELEFONO
Address ESSENTIAL PE_L1_DIRECCION	COLUMN NAME · DATA TYPE · VALUE FORMAT	PE_DIRECCION
Address complement ESSENTIAL PE_L1_DIRECCION_COMPLEMENTO	COLUMN NAME	PE_CM_ALFANUM
Bank account PE_L1_CUENTA_BANCARIA	COLUMN NAME	PE_CM_DIGITOS
Payment card PE_L1_TARJETA	COLUMN NAME · VALUE FORMAT	PE_TARJETA

DATA CLASS	FOUND BY	ALGORITHM
Digital wallet `PE_L1_BILLETERA`	COLUMN NAME · VALUE FORMAT	`PE_CM_ALFANUM`
IP address `PE_L1_IP`	COLUMN NAME · VALUE FORMAT	`PE_IP`
Device identifier `PE_L1_DISPOSITIVO`	COLUMN NAME · VALUE FORMAT	`PE_CM_HEX`
Cookies and tracking identifiers `PE_L1_COOKIE`	COLUMN NAME	`PE_CM_ALFANUM`
Vehicle plate `PE_L1_PLACA`	COLUMN NAME · VALUE FORMAT	`PE_CM_DIGITOS_LETRAS`
Vehicle and licence `PE_L1_VEHICULO`	COLUMN NAME · VALUE FORMAT	`PE_CM_ALFANUM`
Property `PE_L1_INMUEBLE`	COLUMN NAME	`PE_CM_ALFANUM`
Customer and contract numbers `PE_L1_CONTRATO`	COLUMN NAME	`PE_CM_ALFANUM`
User and social media `PE_L1_USUARIO`	COLUMN NAME	`PE_CM_ALFANUM`
Credentials `PE_L1_CREDENCIAL`	COLUMN NAME · VALUE FORMAT	`PE_REDACTAR`
Geolocation `PE_L1_GEOLOCALIZACION`	COLUMN NAME · VALUE FORMAT	`PE_COORDENADA`
Free text `PE_L1_TEXTO_LIBRE`	COLUMN NAME · DATA TYPE · VALUE FORMAT	`PE_TEXTO_LIBRE`
QUASI-IDENTIFIERS		
Birth date ESSENTIAL `PE_L2_FECHA_NACIMIENTO`	COLUMN NAME · DATA TYPE	`PE_FECHA_NACIMIENTO`
Birth year `PE_L2_ANIO_NACIMIENTO`	COLUMN NAME · DATA TYPE	`PE_ANIO`
Age `PE_L2_EDAD`	COLUMN NAME · DATA TYPE	`PE_EDAD`
Personal event dates `PE_L2_FECHA_EVENTO`	COLUMN NAME · DATA TYPE	`PE_FECHA_EVENTO`
Sex `PE_L2_SEXO`	COLUMN NAME · KNOWN VALUES	`PE_SEXO`
Distrito and provincia `PE_L2_DISTRITO`	COLUMN NAME · VALUE FORMAT	`PE_DISTRITO`
UBIGEO `PE_L2_UBIGEO`	COLUMN NAME · VALUE FORMAT	`PE_UBIGEO`
Centro poblado and caserío `PE_L2_CENTRO_POBLADO`	COLUMN NAME	`PE_ZONA_RURAL`
Urbanización and asentamiento humano `PE_L2_URBANIZACION`	COLUMN NAME	`PE_URBANIZACION`
Postal code `PE_L2_CODIGO_POSTAL`	COLUMN NAME · DATA TYPE	`PE_CODIGO_POSTAL`

DATA CLASS	FOUND BY	ALGORITHM
Marital status PE_L2_ESTADO_CIVIL	COLUMN NAME · VALUE FORMAT	PE_ESTADO_CIVIL
Occupation PE_L2_OCUPACION	COLUMN NAME	PE_OCUPACION_O_CODIGO
Employer PE_L2_EMPLEADOR	COLUMN NAME	PE_EMPLEADOR
Education PE_L2_NIVEL_EDUCATIVO	COLUMN NAME · VALUE FORMAT	PE_NIVEL_EDUCATIVO
School PE_L2_INSTITUCION_EDUCATIVA	COLUMN NAME	PE_INSTITUCION_EDUCATIVA
Dependants and children PE_L2_PERSONAS_A_CARGO	COLUMN NAME · DATA TYPE	PE_CONTEO_LIMITADO
SENSITIVE DATA (ART. 2, 5)		
Ethnic origin PE_L3_ORIGEN_ETNICO	COLUMN NAME · VALUE FORMAT	PE_ORIGEN_ETNICO
Indigenous people PE_L3_PUEBLO_INDIGENA	COLUMN NAME · VALUE FORMAT	PE_PUEBLO_INDIGENA
Language PE_L3_LENGUA	COLUMN NAME · VALUE FORMAT	PE_LENGUA
Nationality PE_L3_NACIONALIDAD	COLUMN NAME · KNOWN VALUES	PE_NACIONALIDAD
Religion PE_L3_RELIGION	COLUMN NAME · VALUE FORMAT	PE_RELIGION
Philosophical and moral convictions PE_L3_CONVICCION_FILOSOFICA	COLUMN NAME	PE_CATEGORIA_SUPRIMIDA
Political opinion PE_L3_OPINION_POLITICA	COLUMN NAME · VALUE FORMAT	PE_IDEOLOGIA_POLITICA
Party membership PE_L3_AFILIACION_PARTIDARIA	COLUMN NAME · VALUE FORMAT	PE_PARTIDO
Trade union membership PE_L3_AFILIACION_SINDICAL	COLUMN NAME · VALUE FORMAT	PE_CATEGORIA_SUPRIMIDA
Biometric data PE_L3_BIOMETRICO	COLUMN NAME · DATA TYPE	PE_REDACTAR
Genetic data PE_L3_GENETICO	COLUMN NAME · VALUE FORMAT	PE_REDACTAR
Sexual orientation PE_L3_ORIENTACION_SEXUAL	COLUMN NAME · VALUE FORMAT	PE_ORIENTACION_SEXUAL
Sexual life PE_L3_VIDA_SEXUAL	COLUMN NAME	PE_CATEGORIA_SUPRIMIDA
Gender identity PE_L3_IDENTIDAD_GENERO	COLUMN NAME · VALUE FORMAT	PE_IDENTIDAD_GENERO
Victims PE_L3_VICTIMA	COLUMN NAME	PE_CATEGORIA_SUPRIMIDA
HEALTH DATA		

DATA CLASS	FOUND BY	ALGORITHM
Health identifiers `PE_SALUD_IDENTIFICADOR`	COLUMN NAME	`PE_CM_ALFANUM`
Health insurance `PE_SALUD_COBERTURA`	COLUMN NAME VALUE FORMAT	`PE_COBERTURA_SALUD`
Diagnosis and ICD-10 `PE_SALUD_DIAGNOSTICO`	COLUMN NAME VALUE FORMAT	`PE_DIAGNOSTICO`
Procedures `PE_SALUD_PROCEDIMIENTO`	COLUMN NAME	`PE_PROCEDIMIENTO`
Medicines `PE_SALUD_MEDICAMENTO`	COLUMN NAME VALUE FORMAT	`PE_MEDICAMENTO`
Clinical text `PE_SALUD_TEXTO_CLINICO`	COLUMN NAME DATA TYPE	`PE_TEXTO_LIBRE`
Test results `PE_SALUD_RESULTADO_EXAMEN`	COLUMN NAME	`PE_CM_ALFANUM`
HIV and sexually transmitted infections `PE_SALUD_VIH`	COLUMN NAME	`PE_CATEGORIA_SUPRIMIDA`
Blood group and Rh `PE_SALUD_GRUPO_SANGUINEO`	COLUMN NAME VALUE FORMAT	`dlpx-core:BloodType SL`
Disability `PE_SALUD_DISCAPACIDAD`	COLUMN NAME VALUE FORMAT	`PE_DISCAPACIDAD`
Occupational health `PE_SALUD_OCUPACIONAL`	COLUMN NAME VALUE FORMAT	`PE_APTITUD_LABORAL`
Sexual and reproductive health `PE_SALUD_REPRODUCTIVA`	COLUMN NAME	`PE_CATEGORIA_SUPRIMIDA`
Mental health `PE_SALUD_MENTAL`	COLUMN NAME	`PE_CATEGORIA_SUPRIMIDA`
FINANCIAL AND SOCIOECONOMIC DATA		
Income and pay `PE_FIN_INGRESOS`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Credit history `PE_FIN_HISTORIAL_CREDITICIO`	COLUMN NAME	`PE_HISTORIAL_CREDITICIO`
Debt `PE_FIN_DEUDA`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Assets `PE_FIN_PATRIMONIO`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Pension and benefits `PE_FIN_PENSION`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Social programmes `PE_FIN_PROGRAMA_SOCIAL`	COLUMN NAME VALUE FORMAT	`PE_PROGRAMA_SOCIAL`
SISFOH classification `PE_FIN_CLASIFICACION_SOCIOECONOMICA`	COLUMN NAME VALUE FORMAT	`PE_CLASIFICACION_SOCIOECONOMICA`
Housing tenure `PE_FIN_VIVIENDA`	COLUMN NAME VALUE FORMAT	`PE_TENENCIA_VIVIENDA`
CRIMINAL RECORDS		

DATA CLASS	FOUND BY	ALGORITHM
Criminal records `PE_PENAL_ANTECEDENTES`	COLUMN NAME	`PE_ANTECEDENTES`
Court or prosecution file `PE_PENAL_EXPEDIENTE`	COLUMN NAME VALUE FORMAT	`PE_EXPEDIENTE_JUDICIAL`

PART 8

Direct identifiers

Data that identifies a person on its own. The DNI is the key of almost every Peruvian record, and the RUC of a natural person is the number 10 followed by that same DNI. Both come out with a valid check digit, because applications validate it, and both mask the same way everywhere.

8.1 DNI

PE_L1_DNI

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The number of the Documento Nacional de Identidad issued by RENIEC: eight digits, `45678901`, the key of almost every Peruvian record. The card prints a **verification character** beside it, and systems store it glued to the number (`45678901-7`) or in a column of its own.

The same data class takes RUCs: a document column often holds both.

HOW IT IS FOUND

COLUMN NAME PE_L1_DNI – Path

90%

```
dni d_n_i documento_?nacional(_?de_?identidad)? n(o|ro|um|umero)_?(dni|documento|doc|identificaci[oó]n|id(ent)?) doc_?identidad nro_?doc num_?doc cui c[oó]digo_?[uú]nico_?de_?identificaci[oó]n (cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)_?(dni|doc|documento) dni_?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)
```

60%

```
documento doc identificaci[oó]n
```

DATA TYPE PE_L1_DNI – Type

Accepts text (length 8 or more) or number (length 8 or more); any other type rules the data class out.

VALUE FORMAT PE_L1_DNI – Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{8}[\s\-][0-9A-JK])`	90%	
`0*(?:\d{8})`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Digit by digit, with the verification character recomputed — the digit series or the letter series, whichever the value carries. A value shaped like a RUC is masked as a RUC.

`PE_DOCUMENTO` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

> `PE_RUC` **STRING ALGORITHM CHAIN**
>
> Applies `PE_RUC_CD` → `PE_RUC_ESPACIO` → `PE_RUC_AUX` → `PE_RUC_VERIFICADOR` , in this order.
>
> > `PE_RUC_CD` **CHECK DIGIT**
> >
> > Masks the digits with `PE_RUC_CUERPO` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 2 3 4 5` from the right.
> >
> > > `PE_RUC_CUERPO` **REGEX DECOMPOSE**
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_DIGITOS` .
> > >
> > > > `PE_CM_DIGITOS_ANCHO_FIJO` **CHARACTER MAPPING**
> > > >
> > > > Maps each character to another of its group (1 groups), keeping length and format.
> > > >
> > > > `PE_CM_DIGITOS` **CHARACTER MAPPING**
> > > >
> > > > Maps each character to another of its group (1 groups), keeping length and format.
> >
> > `PE_RUC_ESPACIO` **REGEX DECOMPOSE**
> >
> > Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.
> >
> > `PE_RUC_AUX` **CHECK DIGIT**
> >
> > Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6 8 10` from the right.
> >
> > > `PE_IDENTIDAD` **REGEX DECOMPOSE**
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> >
> > `PE_RUC_VERIFICADOR` **REGEX DECOMPOSE**
> >
> > Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> `PE_DNI_CON_DIGITO` **STRING ALGORITHM CHAIN**
>
> Applies `PE_DNI_CD` → `PE_DNI_ESPACIO` → `PE_DNI_AUX` → `PE_DNI_VERIFICADOR` , in this order.
>
> > `PE_DNI_CD` **CHECK DIGIT**
> >
> > Masks the digits with `PE_CM_DIGITOS_ANCHO_FIJO` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 2 3` from the right.
> >
> > `PE_DNI_ESPACIO` **REGEX DECOMPOSE**
> >
> > Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.
> >
> > `PE_DNI_AUX` **CHECK DIGIT**
> >
> > Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6` from the right.
> >
> > `PE_DNI_VERIFICADOR` **REGEX DECOMPOSE**
> >
> > Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> `PE_DNI_CON_LETRA` **STRING ALGORITHM CHAIN**
>
> Applies `PE_DNI_CD` → `PE_DNI_ESPACIO` → `PE_DNI_AUX` → `PE_DNI_LETRA` , in this order.
>
> > `PE_DNI_LETRA` **REGEX DECOMPOSE**
> >
> > Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> `PE_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
45678901	→	**32016288**
45678901-7	→	**32016288-8**
20131378972	→	**20817541918**

8.2 RUC

PE_L1_RUC

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The eleven-digit taxpayer number of SUNAT, `20131378972` : two digits for the kind of taxpayer — **10** and 15 for a natural person, 17 for an estate, 20 for a legal person —, eight for the number and one check digit. For a natural person with a 10, those eight digits **are the DNI**.

HOW IT IS FOUND

COLUMN NAME PE_L1_RUC — Path

90%

```
ruc[a-z0-9_]*  n(o|ro|um|umero)_?ruc  registro_?[uú]nico_?(de_?)?contribuyente
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|
trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietar
io|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elec
tor|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionis
ta|aportante|derechohabiente)_?ruc  ruc_?
(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|t
rabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietari
o|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elect
or|votante|contribuyente|proveedor|socio|conyuge|padre|madre|hijo|familiar|jubilado|pensionist
a|aportante|derechohabiente)  id_?tributario  tax_?id
```

VALUE FORMAT PE_L1_RUC — Regex

PATTERN	STRENGTH	NOTES
`0*(?:(10|15|17|20)\d{9})`	90%	
`0*(?:\d{11})`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The two digits of the kind of taxpayer stay, the eight of the number are mapped and the check digit is recomputed, so DNI and RUC of the same person keep agreeing.

`PE_RUC_COLUMNA` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_RUC` .

> `PE_RUC` STRING ALGORITHM CHAIN
>
> Applies `PE_RUC_CD` → `PE_RUC_ESPACIO` → `PE_RUC_AUX` → `PE_RUC_VERIFICADOR` , in this order.
>
> > `PE_RUC_CD` CHECK DIGIT
> >
> > Masks the digits with `PE_RUC_CUERPO` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 2 3 4 5` from the right.
> >
> > > `PE_RUC_CUERPO` REGEX DECOMPOSE
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_DIGITOS` .
> > >
> > > > `PE_CM_DIGITOS_ANCHO_FIJO` CHARACTER MAPPING
> > > >
> > > > Maps each character to another of its group (1 groups), keeping length and format.
> > > >
> > > > `PE_CM_DIGITOS` CHARACTER MAPPING
> > > >
> > > > Maps each character to another of its group (1 groups), keeping length and format.
> >
> > `PE_RUC_ESPACIO` REGEX DECOMPOSE
> >
> > Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.
> >
> > `PE_RUC_AUX` CHECK DIGIT

Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6 8 10` from the right.

`PE_IDENTIDAD` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`PE_RUC_VERIFICADOR` REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
20131378972	→	**20817541918**
10456789017	→	**10320162888**

8.3 Verification character

PE_L1_DIGITO_VERIFICACION

COLUMN NAME · DATA TYPE · CHARACTER MAPPING

WHAT IT IS

The DNI's verification digit or letter kept in a column of its own.

HOW IT IS FOUND

COLUMN NAME PE_L1_DIGITO_VERIFICACION — Path

80% `d[ií]gito_?(de_?)?verificaci[oó]n` `digito_?verificador` `dig_?ver` `dv_?dni` `dv` `c[oó]digo_?(de_?)?verificaci[oó]n_?(dni|documento)` `check_?digit`

DATA TYPE PE_L1_DIGITO_VERIFICACION — Type

Accepts text (length up to 1) or number (length up to 1); any other type rules the data class out.

HOW IT IS MASKED

Digits to digits and letters to letters. It cannot be recomputed there: the number it belongs to is not in the value.

`PE_CM_DIGITOS_LETRAS` CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
7	→	**5**
K	→	**0**

8.4 Foreigner documents

PE_L1_CARNE_EXTRANJERIA

ESSENTIAL · COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

The carné de extranjería, the PTP and CPP permits of Migraciones, the migratory status and the visa number — the documents of the Venezuelan and other foreign communities.

HOW IT IS FOUND

COLUMN NAME PE_L1_CARNE_EXTRANJERIA — Path

85% `carn[eé]_?(de_?)?extranjer[ií]a` `carnet_?extranjeria` `c[\.\s]?e[\.\s]?_?(n(o|ro|um|umero))?` `n(o|ro|um|umero)_?(carn[eé]|extranjer[ií]a)` `permiso_?temporal_?(de_?)?permanencia` `ptp` `carn[eé]_?(de_?)?permiso_?temporal` `cpp` `calidad_?migratoria` `condici[oó]n_?migratoria` `n(o|ro|um|umero)_?(de_?)?(residencia|migraciones)` `visa(_?n(o|ro|um|umero))?`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
001234567	→	**005649285**

8.5 Passport

PE_L1_PASAPORTE

ESSENTIAL COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

Peruvian passport numbers and foreign ones.

HOW IT IS FOUND

COLUMN NAME PE_L1_PASAPORTE — Path

90% `pasaporte[a-z0-9_]*` `n(o|ro|um|umero)_?pasaporte` `passport[a-z0-9_]*`

VALUE FORMAT PE_L1_PASAPORTE — Regex

PATTERN	STRENGTH	NOTES
`[A-Z]{1,2}\d{6,8}`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
A1234567	→	**U2438404**

8.6 Civil registry records

PE_L1_ACTA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The acta, book and folio of the birth, marriage and death records RENIEC and the municipal registry offices keep, and the CUA that numbers them.

HOW IT IS FOUND

COLUMN NAME PE_L1_ACTA — Path

85% `acta_?(de_?)?(nacimiento|matrimonio|defunci[oó]n)(_?(n(o|ro|um|umero)|folio|libro))?` `n(o|ro|um|umero)_?acta` `partida_?(de_?)?(nacimiento|matrimonio|defunci[oó]n)` `folio_?(acta|libro|partida)` `libro_?(acta|registro)` `asiento_?registral_?civil` `c[oó]digo_?[uú]nico_?de_?acta` `cua`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
123/1985	→	**091/7273**

8.7 EsSalud, AFP and ONP

PE_L1_SEGURO_SOCIAL

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The EsSalud *autogenerado*, the CUSPP of the private pension system and the ONP and individual account numbers.

HOW IT IS FOUND

COLUMN NAME PE_L1_SEGURO_SOCIAL — Path

85% `autogenerado` `c[oó]digo_?(de_?)?(asegurado|autogenerado|essalud)` `n(o|ro|um|umero)_?(essalud|asegurado|afiliado|autogenerado)` `essalud[a-z0-9_]*` `cuspp` `c[oó]digo_?[uú]nico_?(de_?)?identificaci[oó]n_?(del_?)?spp` `n(o|ro|um|umero)_?(afp|onp|cuenta_?individual|cic)` `afp[a-z0-9_]*` `onp[a-z0-9_]*` `n(o|ro|um|umero)_?(de_?)?aportante`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
JUQU12345678	→	**SAGI03506074**

8.8 Student code

PE_L1_CODIGO_ESTUDIANTE

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The student code, the enrolment number and the university card, and the SIAGIE identifier of school pupils.

HOW IT IS FOUND

COLUMN NAME PE_L1_CODIGO_ESTUDIANTE — Path

80% `c[oó]digo_?(de_?)?(estudiante|alumno|matr[ií]cula|educando)` `cod_?(alumno|estudiante)` `n(o|ro|um|umero)_?(de_?)?(matr[ií]cula|carn[eé]_?universitario)` `carn[eé]_?(de_?)?(estudiante|universitario)` `id_?(alumno|estudiante)` `c[oó]digo_?modular_?(del_?)?estudiante` `siagie`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
20201234	→	**10992819**

8.9 Professional registration

PE_L1_COLEGIATURA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The number of the professional college — CMP for physicians, CIP for engineers, CAL for lawyers —, published together with the name.

HOW IT IS FOUND

COLUMN NAME PE_L1_COLEGIATURA — Path

85% `colegiatura` `n(o|ro|um|umero)_?(de_?)?(colegiatura|colegiado|registro_?profesional)` `cmp` `c[oó]digo_?(m[eé]dico|profesional)` `registro_?(m[eé]dico|profesional|cip|cal)` `cip` `matr[ií]cula_?profesional` `habilitaci[oó]n_?profesional`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
CMP-12345	→	**DRF-17346**

8.10 Given name

PE_L1_NOMBRE

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The given name, one or two: `Milagros`, `Juan Carlos`, `María del Rosario`.

HOW IT IS FOUND

COLUMN NAME PE_L1_NOMBRE – Path

85% `primer_?nombre` `segundo_?nombre` `nombres?_?(de_?)?pila` `pri(mer)?_?nom` `seg(undo)?_?nom` `nombre_?1` `nombre_?2` `prenombres?` `first_?names?` `given_?names?` `fname` `middle_?names?` `nombres` *(except when followed by `completos?`, `y_?apellidos?`, `apellidos?`)*

50% `nombre` *(except when followed by list E1)* `nom` *(except when followed by list E2)* `name` *(except when followed by list E3)*

KNOWN VALUES PE_L1_NOMBRE – List

LIST	VALUES	STRENGTH	FIRST VALUES
`pe-detectar-nombres.txt`	456	70%	José, Jose, María, Maria, Juan, Carlos

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Word by word, replacing each name with one from the list. `de`, `del` and `la` stay in place. More than four words are masked character by character.

`PE_NOMBRES` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

`PE_PALABRA_NOMBRE` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_NOMBRE`.

`PE_NOMBRE` NAME

Replaces the name with one of the 395 names in `pe-nombres.txt`.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
María del Rosario	→	**Paulino del Javier**
JUAN CARLOS	→	**FILOMENO BRENDA**

8.11 Surname

PE_L1_APELLIDO

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The surnames — the father's and the mother's —, in separate columns or together: `Quispe Huamán`. Quispe and Mamani, the two most common, are Quechua and Aymara.

HOW IT IS FOUND

COLUMN NAME PE_L1_APELLIDO – Path

85% `apellidos?` `primer_?apellido` `segundo_?apellido` `apellido_?(1|2|paterno|materno|casada)` `ape_?(1|2|pat|mat)` `ap_?(paterno|materno)` `last_?names?` `surnames?` `family_?names?` `lname`

KNOWN VALUES PE_L1_APELLIDO — List

LIST	VALUES	STRENGTH	FIRST VALUES
pe-detectar-apellidos.txt	519	70%	Quispe, Mamani, Flores, Huamán, Huaman, Sánchez

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Word by word, replacing each surname with one from the list; particles stay in place.

PE_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE_CM_ALFANUM.

> PE_PALABRA_APELLIDO REGEX DECOMPOSE
>
> Matches one pattern and gives each part of the value its own action; when none matches: PE_APELLIDO.
>
> > PE_APELLIDO NAME
> >
> > Replaces the name with one of the 422 names in pe-apellidos.txt.
>
> PE_CM_ALFANUM CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
Quispe Huamán	→	**Ortiz Zambrano**

8.12 Full name

PE_L1_NOMBRE_COMPLETO

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Given names and surnames in one column: Juan Carlos Quispe Huamán. Also those of the mother, father, guardian, spouse, partner, emergency contact and reference.

RENIEC lists write APELLIDOS, NOMBRES with a comma; the policy reads that form too.

HOW IT IS FOUND

COLUMN NAME PE_L1_NOMBRE_COMPLETO — Path

90% `nombres?_?(completos?|y_?apellidos?|apellidos?)` `apellidos?_?(y_?)?nombres?` `nombre_?(del_?|de_?la_?)?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|imputado|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|imputado|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)_?(nombre|nom)` `nom_?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|fiador|avalista|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|imputado|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `nombre_?(madre|padre|tutor|conyuge|conviviente|contacto_?emergencia|referencia)` `full_?name` `person_?name` `razon_?social_?persona_?natural`

60% `madre` `padre` `tutor` `conyuge` `conviviente` `representante_?legal` `apoderado` `fiador` `beneficiario` `heredero` `titular` `referencia_?(personal|familiar)`

50% `nombre` *(except when followed by list E1)* `nom` *(except when followed by list E2)* `name` *(except when followed by list E3)*

KNOWN VALUES PE_L1_NOMBRE_COMPLETO — List

LIST	VALUES	STRENGTH	FIRST VALUES
pe-detectar-nombres.txt	456	65%	José, Jose, María, Maria, Juan, Carlos
pe-detectar-apellidos.txt	519	65%	Quispe, Mamani, Flores, Huamán, Huaman, Sánchez

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

In Peruvian order: two words are a given name and a surname; three, a given name and two surnames; four, two and two. The comma form is masked keeping the comma.

`PE_NOMBRE_COMPLETO` **REGEX DECOMPOSE**

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

`PE_APELLIDOS` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

`PE_PALABRA_APELLIDO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_APELLIDO`.

`PE_APELLIDO` NAME

Replaces the name with one of the 422 names in `pe-apellidos.txt`.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_NOMBRES` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

`PE_PALABRA_NOMBRE` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_NOMBRE`.

`PE_NOMBRE` NAME

Replaces the name with one of the 395 names in `pe-nombres.txt`.

EXAMPLES

VALUE		MASKED
Juan Carlos Quispe Huamán	→	**Filomeno Brenda Ortiz Zambrano**
Milagros Quispe	→	**Pablo Ortiz**
QUISPE HUAMÁN, JUAN CARLOS	→	**ORTIZ ZAMBRANO, FILOMENO BRENDA**

8.13 Business name

PE_L1_RAZON_SOCIAL

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The business name of a sole trader, beside the RUC that names them. The law protects the natural person behind it; a company as such is not a data subject.

HOW IT IS FOUND

COLUMN NAME PE_L1_RAZON_SOCIAL — Path

85% `raz[oó]n_?social` `razonsocial` `nombre_?(comercial|de_?la_?empresa|empresa)` `denominaci[oó]n_?social` `nombre_?legal` `business_?name` `company_?name` `legal_?name`

HOW IT IS MASKED

Replaced by a fictitious business name.

`PE_RAZON_SOCIAL` SECURE LOOKUP

Replaces the value with one of the 20 values in `pe-razones-sociales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Corporación Comercial Peruana S.A.C.	→	**Café de Altura Chanchamayo S.A.C.**

8.14 E-mail

PE_L1_EMAIL

ESSENTIAL · COLUMN NAME · DATA TYPE · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Personal and corporate e-mail addresses.

HOW IT IS FOUND

COLUMN NAME PE_L1_EMAIL — Path

85% `e_?mail[a-z0-9_]*` `mail[a-z0-9_]*` `correo(_?electr[oó]nico)?[a-z0-9_]*` `dir_?correo` `direcci[oó]n_?(de_?)?correo` `direcci[oó]n_?electr[oó]nica`

DATA TYPE PE_L1_EMAIL — Type

Accepts text (length 5 or more); any other type rules the data class out.

VALUE FORMAT PE_L1_EMAIL — Regex

PATTERN	STRENGTH	NOTES
`[\p{L}\p{N}._%+'\-]+@[\p{L}\p{N}.\-]+\.[A-Za-z]{2,24}`	100%	

A value that matches none counts 20% against the data class.

HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — `.com`, `.com.pe` — becomes `.test`, reserved for tests: a masked address can never deliver.

`PE_EMAIL` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`milagros.quispe@gmail.com`	→	**`gevevjew.cuahxa@pgoeb.test`**

8.15 Phone

PE_L1_TELEFONO

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Nine-digit mobiles, all starting with 9, and landlines with an area code — 1 for Lima and Callao, two digits elsewhere — and six or seven digits.

HOW IT IS FOUND

COLUMN NAME PE_L1_TELEFONO — Path

85% `tel` `tel[eé]fono[a-z0-9_]*` `tel_?(fijo|casa|oficina|trabajo|contacto|celular|m[oó]vil)` `celular[a-z0-9_]*` `cel` `m[oó]vil` `whats_?app` `anexo_?telef[oó]nico` `n(o|ro|um|umero)_?(tel|tel[eé]fono|celular|cel|contacto)` `fax` `phone[a-z0-9_]*` `mobile[a-z0-9_]*`

VALUE FORMAT PE_L1_TELEFONO — Regex

PATTERN	STRENGTH	NOTES
`0*(?:(\+?51[\s\-]?)?9\d{2}[\s\-]?\d{3}[\s\-]?\d{3})`	90%	
`\(?0?1\)?[\s\-]?\d{3}[\s\-]?\d{4}`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

`+51`, the leading 9 of a mobile and the area code of a landline stay; the rest of the digits are mapped, so the number keeps saying mobile or landline and which region.

`PE_TELEFONO` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_DIGITOS`.

> `PE_CM_DIGITOS` **CHARACTER MAPPING**
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
+51 987 654 321	→	**+51 912 410 723**
(01) 234-5678	→	**(01) 158-7565**

8.16 Address

`PE_L1_DIRECCION`

ESSENTIAL · COLUMN NAME · DATA TYPE · VALUE FORMAT · SECURE LOOKUP

WHAT IT IS

The address with the kind of street and the number — `Av. Arequipa 1234, Urb. Santa Patricia` —, and the `Mz. B Lt. 12` of an asentamiento humano.

HOW IT IS FOUND

> **COLUMN NAME** `PE_L1_DIRECCION – Path`
>
> **80%** `direcci[oó]n[a-z0-9_]*` *(except when preceded by* `mac`, `ip`, `e_?mail`, `web`, `url`, `correo`*; except when followed by list E4)* `domicilio[a-z0-9_]*` `dom_?(residencia|particular|laboral|cliente|fiscal)` `residencia` `lugar_?(de_?)?residencia` `direccion_?fiscal` `calle(_?y_?n[uú]mero)?` `address[a-z0-9_]*` *(except when preceded by* `mac`, `ip`, `e_?mail`, `web`, `url`*; except when followed by list E5)*

> **DATA TYPE** `PE_L1_DIRECCION – Type`
>
> Accepts text (length 6 or more); any other type rules the data class out.

> **VALUE FORMAT** `PE_L1_DIRECCION – Regex`
>
> | PATTERN | STRENGTH | NOTES |
> |---|---|---|
> | `(?i)(av\|avenida\|jr\|jir[oó]n\|calle\|ca\|pasaje\|psje\|prolongaci[oó]n\|alameda\|malec[oó]n\|carretera)\.?\s+[a-záéíóúñ0-9.' ]{2,40}\s*(n°\|nro\.?\|#)?\s*\d{1,5}.*` | 80% | |
> | `(?i).*\b(mz\|manzana)\.?\s*[a-z0-9]{1,3}\s*(lt\|lote)\.?\s*\d{1,3}.*` | 80% | |
> | `(?i).*\b(urb\|urbanizaci[oó]n\|aa\.?hh\.?\|asentamiento_?humano\|pueblo_?joven\|coop\|cooperativa\|residencial\|conjunto_?habitacional)\.?\s+[a-záéíóúñ0-9.' ]{2,40}.*` | 70% | |
> | `(?i)k(m\|il[oó]metro)\.?\s*\d{1,3}\s+(carretera\|autopista\|v[ií]a).*` | 70% | |
>
> A value that matches none counts 20% against the data class.

HOW IT IS MASKED

The whole line is replaced by a fictitious Peruvian address of the same form. Urbanización, centro poblado, distrito and postal code have data classes of their own for when they live in separate columns.

`PE_DIRECCION` **SECURE LOOKUP**

Replaces the value with one of the 4,000 values in `pe-direcciones.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Av. Arequipa 1234, Urb. Santa Patricia`	→	**`Res. Santa Cruz, Carretera Independencia 446`**

8.17 Address complement

PE_L1_DIRECCION_COMPLEMENTO

ESSENTIAL · COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

Manzana, lote, interior, flat, floor, block or stage in columns of their own.

HOW IT IS FOUND

COLUMN NAME — PE_L1_DIRECCION_COMPLEMENTO — Path

70% `manzana` `mz` `lote` `lt` `interior` `int_?(direccion|domicilio)` `dpto_?(n(o|ro|um|umero))?` `departamento_?(n(o|ro|um|umero)|interior)` `piso` `block` `bloque` `etapa` `sector_?(vivienda|domicilio)` `n(o|ro|um|umero)_?(casa|puerta|dpto|interior)` `complemento(_?direcci[oó]n)?` `referencia_?(de_?)?(direcci[oó]n|domicilio)`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`Mz. B Lt. 12`	→	**`Rt. Z Kj. 69`**

8.18 Bank account

PE_L1_CUENTA_BANCARIA

COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

Savings, current, payroll, CTS and detraction account numbers, and the twenty-digit CCI. The bank code is not personal data and has no data class.

HOW IT IS FOUND

COLUMN NAME — PE_L1_CUENTA_BANCARIA — Path

85% `cuenta_?(bancaria|banco|corriente|ahorro|sueldo|haberes|cts|abono|dep[oó]sito|destino|origen|detracci[oó]n)` `n(o|ro|um|umero)_?(de_?)?(cuenta|cta)` `nro_?cta` `num_?cta` `cta_?(cte|corriente|ahorro|bancaria)` `cci` `c[oó]digo_?(de_?)?cuenta_?interbancario` `iban` `account_?(no|num|number)` `bank_?account`

HOW IT IS MASKED

Digits are mapped; separators stay.

`PE_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
0011-0123-0100123456	→	**0062-7186-9933536389**

8.19 Payment card

PE_L1_TARJETA

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

HOW IT IS FOUND

COLUMN NAME PE_L1_TARJETA – Path

85% `tarjeta(_?(cr[eé]dito|d[eé]bito|n(o|ro|um|umero)))?` `n(o|ro|um|umero)_?tarjeta` `pan` `card_?(no|num|number)` `credit_?card` `tc_?(numero|num|nro)`

VALUE FORMAT PE_L1_TARJETA – Regex

PATTERN	STRENGTH	NOTES
`0*(?:3[47]\d{13}|[1-9]\d{12,13}|[1-9]\d{15,18})`	90%	LUHN check digit; after removing `[\s\-]`

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Credit Card, which comes with Delphix: another number that passes the Luhn check, keeping the first six digits — the issuer, not the holder. Only values shaped like a card reach it; the others pass unchanged instead of failing the row.

`PE_TARJETA` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `dlpx-core:Credit Card` PAYMENT CARD
>
> Comes with Delphix and is used as it ships. Replaces the card number with another that passes the Luhn check, keeping the first six digits — the issuer, not the holder.

EXAMPLES

VALUE		MASKED
4557 8800 0000 0001	→	**4557 8886 1298 3134**

8.20 Digital wallet

PE_L1_BILLETERA

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

Yape, Plin and other mobile wallet identifiers — keyed by the phone number — and the bitcoin addresses some systems keep.

HOW IT IS FOUND

COLUMN NAME PE_L1_BILLETERA – Path

85% `billetera(_?(digital|electr[oó]nica|m[oó]vil))?` `wallet(_?(id|address|direccion))?` `yape[a-z0-9_]*` `plin[a-z0-9_]*` `tunki` `lukita` `agora_?pay` `c[oó]digo_?qr` `qr_?(id|codigo)` `direcci[oó]n_?(bitcoin|btc|wallet)` `btc_?(address|direccion)`

VALUE FORMAT PE_L1_BILLETERA — Regex

PATTERN	STRENGTH	NOTES
`(bc1|[13])[a-zA-HJ-NP-Z0-9]{25,59}`	70%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping the shape. A masked bitcoin address no longer passes its own checksum, which is what a test copy wants.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`bc1qar0srrr7xfkvy5l643lydnw9re59gtzzwf5mdq`	→	**`mh2wis8tsdm0pdjdv8v511yqxqg4ze53qkjbvp7dgf`**

8.21 IP address

PE_L1_IP

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses from access logs and transactions.

HOW IT IS FOUND

COLUMN NAME PE_L1_IP — Path

85% `ip` `ip_?(address|addr|origen|destino|cliente|usuario|acceso|login|remota|publica)` `direcci[oó]n_?ip` `dir_?ip` `ipv[46]` `remote_?addr(ess)?` `client_?ip` `host_?ip`

VALUE FORMAT PE_L1_IP — Regex

PATTERN	STRENGTH	NOTES
`0*(?:((25[0-5]|2[0-4]\d|1?\d?\d)\.){3}(25[0-5]|2[0-4]\d|1?\d?\d))`	100%	
`0*(?:[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7})`	90%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each IPv4 octet is mapped to 1–254; an IPv6 address is mapped keeping its hex characters.

`PE_IP` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_HEX`.

> `PE_OCTETO` NUMERIC MAPPING
>
> Maps the number to another between 1 and 254.
>
> `PE_CM_HEX` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`190.117.45.12`	→	**`164.108.51.94`**

8.22 Device identifier

PE_L1_DISPOSITIVO

COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING

WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising and session identifiers.

HOW IT IS FOUND

COLUMN NAME — PE_L1_DISPOSITIVO — Path

80% `imei` `imsi` `iccid` `mac` `mac_?address` `direcci[oó]n_?mac` `device_?(id|uuid|serial)` `id_?(dispositivo|equipo|celular)` `serial_?(equipo|celular|dispositivo)` `advertising_?id` `idfa` `gaid` `session_?id` `id_?sesi[oó]n`

VALUE FORMAT — PE_L1_DISPOSITIVO — Regex

PATTERN	STRENGTH	NOTES
`0*(?:([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2})`	100%	
`0*(?:\d{15})`	80%	LUHN check digit

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, with hex digits kept hex: a MAC stays a MAC.

`PE_CM_HEX` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`3c:22:fb:9a:10:e4`	→	**`5e:83:bb:3c:31:f9`**

8.23 Cookies and tracking identifiers

PE_L1_COOKIE

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

The identifier a site stores in the browser and the campaign parameters that travel with it. It is personal data when it singles out a person through means that can reasonably be used.

HOW IT IS FOUND

COLUMN NAME — PE_L1_COOKIE — Path

80% `cookie(_?(id|value|valor|nombre))?` `cookies` `_?ga` `_?gid` `fbp` `fbclid` `utm_?(source|medium|campaign|term|content)` `id_?(navegador|browser|visitante)` `visitor_?id` `tracking_?id` `client_?id`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
GA1.2.123456789.1700000000	→	**ZE0.5.891552838.7889916184**

8.24 Vehicle plate

PE_L1_PLACA

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

The Placa Única Nacional de Rodaje: three letters and three digits since 2011 (`ABC-123`), a variant that mixes them (`A1B-234`), and two letters and four digits on a motorcycle.

HOW IT IS FOUND

COLUMN NAME — PE_L1_PLACA — Path

85% `placa(_?(veh[ií]culo|carro|moto|automotor|rodaje))?` `n(o|ro|um|umero)_?placa` `placa_?[uú]nica_?nacional_?(de_?)?rodaje` `punr` `matr[ií]cula_?(veh[ií]culo|automotor)` `license_?plate` `plate(_?(no|number))?`

VALUE FORMAT — PE_L1_PLACA — Regex

PATTERN	STRENGTH	NOTES
`[A-Z][0-9A-Z][A-Z][\s\-]?\d{3}`	80%	
`[A-Z]{2}[\s\-]?\d{4}`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Letters to letters and digits to digits, position by position: the plate format stays.

`PE_CM_DIGITOS_LETRAS` CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
ABC-123	→	**AFJ-775**
A1B-234	→	**V9D-496**

8.25 Vehicle and licence

PE_L1_VEHICULO

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

The VIN or chassis number, the engine number, the vehicle identification card and the driving licence — the *brevete*.

HOW IT IS FOUND

COLUMN NAME — PE_L1_VEHICULO — Path

85% `vin` `chasis` `n(o|ro|um|umero)_?(chasis|motor|serie|vin)` `motor_?n(o|ro|um|umero)` `tarjeta_?(de_?)?(identificaci[oó]n_?vehicular|propiedad|circulaci[oó]n)` `licencia_?(de_?)?conducir(_?(n(o|ro|um|umero)))?` `n(o|ro|um|umero)_?licencia` `brevete`

VALUE FORMAT PE_L1_VEHICULO — Regex

PATTERN	STRENGTH	NOTES
`[A-HJ-NPR-Z0-9]{17}`	70%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
8AJHA8CD3J1234567	→	**8EWKE1QN1T0772894**

8.26 Property

PE_L1_INMUEBLE

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The partida registral and ficha of SUNARP, the cadastral code and the self-assessment record: they lead to the home and its owner.

HOW IT IS FOUND

COLUMN NAME PE_L1_INMUEBLE — Path

80% `partida_?(registral|electr[oó]nica)` `n(o|ro|um|umero)_?partida_?registral` `ficha_?registral` `tomo_?(y_?)?folio` `n(o|ro|um|umero)_?(de_?)?(predio|inmueble|catastral)` `c[oó]digo_?(catastral|predial|de_?predio)` `catastro` `cup` `asiento_?registral` `sunarp` `autoval[uú]o_?(n(o|ro|um|umero)|c[oó]digo)` `hr_?pu`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
P12345678	→	**T33398422**

8.27 Customer and contract numbers

PE_L1_CONTRATO

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Contract, policy, loan, application, member, subscriber, supply, meter, case or administrative file numbers, and customer or employee codes.

HOW IT IS FOUND

COLUMN NAME PE_L1_CONTRATO – Path

70% `n(o|ro|um|umero)_?(contrato|p[oó]liza|cr[eé]dito|pr[eé]stamo|solicitud|expediente_?(administrativo|interno)|tr[aá]mite|referencia|cliente|socio|suscriptor|suministro|medidor|servicio|caso|ticket|beneficio)` `contrato_?(n(o|ro|um|umero))` `c[oó]digo_?(cliente|socio|empleado|afiliado|suministro)` `id_?(cliente|socio|empleado|afiliado)` `n(o|ro)_?empleado` `c[oó]digo_?(de_?)?(empleado|trabajador|planilla)` `n(o|ro|um|umero)_?expediente_?siga`

HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
CT-123456	→	**NG-557411**

8.28 User and social media

PE_L1_USUARIO

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Login, alias and social media profiles.

HOW IT IS FOUND

COLUMN NAME PE_L1_USUARIO – Path

70% `usuario` `user_?name` `username` `login` `alias` `apodo` `nick(_?name)?` `perfil_?(instagram|facebook|twitter|tiktok|linkedin|x)` `instagram` `facebook` `twitter` `tiktok` `linkedin` `red_?social` `handle`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
@milagros_quispe	→	**@gevevjew_cuahxa**

8.29 Credentials

PE_L1_CREDENCIAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Passwords and their hashes, PIN, security and OTP codes, CVV, tokens, API keys, the SUNAT *clave SOL*, private keys and seed phrases.

HOW IT IS FOUND

COLUMN NAME PE_L1_CREDENCIAL — Path

90% `contrase[nñ]a` `clave` *(except when followed by* `catastral`, `primaria`, `for[aá]nea`, `valor`, `producto`*)* `password` `passwd` `pwd` `pass` `hash_?(clave|contrasena|password)` `pin` `c[oó]digo_?(seguridad|verificaci[oó]n|acceso|otp)` `cvv` `cvc` `token(_?(acceso|refresh|api))?` `access_?token` `refresh_?token` `api_?key` `llave_?(api|privada)` `secreto` `secret` `otp` `clave_?sol` `clave_?(de_?)?internet` `pregunta_?secreta` `respuesta_?secreta` `frase_?semilla` `seed_?phrase`

VALUE FORMAT PE_L1_CREDENCIAL — Regex

PATTERN	STRENGTH	NOTES
`\$2[abxy]?\$\d{2}\$[./A-Za-z0-9]{53}`	95%	
`\$(1|5|6)\$(rounds=\d+\$)?[./A-Za-z0-9]{1,16}\$[./A-Za-z0-9]{22,86}`	95%	
`\$argon2(id|i|d)\$v=\d+\$m=\d+,t=\d+,p=\d+\$\S+`	95%	
`pbkdf2_sha(1|256|512)\$\d+\$\S+\$\S+`	95%	
`\{(SSHA|SHA|SSHA256|SSHA512|SMD5|MD5|CRYPT)\}\S+`	95%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`. A credential has no use in a test copy, and a hash masked character by character could still be cracked.

`PE_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
`Clave#2024`	→	XXXXXXXXXX

8.30 Geolocation

PE_L1_GEOLOCALIZACION

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, visits, addresses and deliveries.

HOW IT IS FOUND

COLUMN NAME PE_L1_GEOLOCALIZACION — Path

85% `lat` `latitud` `lon` `lng` `longitud` `coordenadas?` `coord_?[xy]` `geo_?(lat|lon|lng|loc|localizacion|posicion)` `georreferenciaci[oó]n` `geolocalizaci[oó]n` `gps(_?(ubicaci[oó]n|posici[oó]n|coordenadas))?` `ubicaci[oó]n_?gps`

50% `long`

VALUE FORMAT PE_L1_GEOLOCALIZACION — Regex

PATTERN	STRENGTH	NOTES
`-\d{1,2}\.\d{3,}\s*,\s*-(7|8)\d\.\d{3,}`	90%	
`-(7[0-9]|8[01])\.\d{4,}`	50%	
`-\d{1,2}\.\d{4,}`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved.

`PE_COORDENADA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

> `PE_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`-12.046374,-77.042793`	→	**`-12.061019,-77.036229`**

8.31 Free text

PE_L1_TEXTO_LIBRE

COLUMN NAME · DATA TYPE · VALUE FORMAT · FREE TEXT REDACTION

WHAT IT IS

Remarks, complaint descriptions, accounts of events and messages where anything gets written — DNI, RUC, phone and names in the middle of a sentence.

HOW IT IS FOUND

COLUMN NAME PE_L1_TEXTO_LIBRE — Path

60% `observaci[oó]n(es)?` `obs` `comentarios?` `notas?` `anotaci[oó]n(es)?` `glosa` `sustento` `descripci[oó]n_?(queja|reclamo|solicitud|caso|hechos|novedad|atenci[oó]n|denuncia)` `detalle_?(reclamo|solicitud|caso|atenci[oó]n)` `hechos` `relato` `justificaci[oó]n` `mensaje` *(except when preceded by list E6; except when followed by list E7)* `texto(_?libre)?` `resumen_?(caso|atenci[oó]n)` `queja` `reclamo` `denuncia_?texto` `remarks?` `comments?` `notes?` `free_?text` `memo`

DATA TYPE PE_L1_TEXTO_LIBRE — Type

Accepts text (length 30 or more); any other type rules the data class out.

VALUE FORMAT PE_L1_TEXTO_LIBRE — Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{8}-[0-9A-JK])`	70%	anywhere in the value
`[\w.+\-]+@[\w\-]+\.[A-Za-z]{2,}`	70%	anywhere in the value
`0*(?:9\d{2}[\s\-]?\d{3}[\s\-]?\d{3})`	60%	anywhere in the value

A value that matches none does not count against the data class.

HOW IT IS MASKED

Free Text Redaction: DNIs, RUCs, e-mail, mobiles, cards, plates and IPs become `[DATO]`, and the names of the lists `[NOMBRE]`.

`PE_TEXTO_LIBRE` **FREE TEXT REDACTION**

Replaces the text matching 8 patterns with `[DATO]`, and the 1,906 terms of `pe-texto-nombres.txt` with `[NOMBRE]`.

EXAMPLES

VALUE		MASKED
`Cliente Quispe, DNI 45678901, correo m.quispe@gmail.com, cel 987654321.`	→	**`Cliente [NOMBRE], DNI [DATO], correo [DATO], cel [DATO].`**

PART 9

Quasi-identifiers

Data that identifies nobody on its own, but does in combination. Peru is divided into 1,892 distritos, and they are small: 1,633 of them had fewer than 20,000 inhabitants in the 2017 census, and hundreds fewer than a thousand. A birth date, a sex and one of those distritos point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

9.1 Birth date

PE_L2_FECHA_NACIMIENTO

ESSENTIAL · COLUMN NAME · DATA TYPE · DATE SHIFT

WHAT IT IS

The birth date, as a date, timestamp or text.

HOW IT IS FOUND

COLUMN NAME PE_L2_FECHA_NACIMIENTO – Path

90% `(fecha|fec|fch|f)_?(de_?)?nac(imiento)?` `fnac` `fechanac(imiento)?` `cumplea[ñn]os` `dob` `date_?of_?birth` `birth_?date` `birthday`

DATA TYPE PE_L2_FECHA_NACIMIENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth date always moves the same way.

`PE_FECHA_NACIMIENTO` DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

VALUE		MASKED
1985-07-20	→	**1985-08-05T00:00**

9.2 Birth year

PE_L2_ANIO_NACIMIENTO

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

WHAT IT IS

The birth year in a column of its own.

HOW IT IS FOUND

COLUMN NAME PE_L2_ANIO_NACIMIENTO – Path

90% `a[ñn]o_?(de_?)?nac(imiento)?` `anio_?nac(imiento)?` `year_?of_?birth` `birth_?year` `yob`

DATA TYPE PE_L2_ANIO_NACIMIENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

HOW IT IS MASKED

The decade stays and the last digit is mapped: `1985` becomes a year from 1980 to 1989.

`PE_ANIO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `PE_CM_DIGITOS` CHARACTER MAPPING
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1985	→	**1980**

9.3 Age

PE_L2_EDAD

COLUMN NAME | DATA TYPE | REGEX DECOMPOSE

WHAT IT IS

Age in years, also age group or five-year band.

HOW IT IS FOUND

> COLUMN NAME PE_L2_EDAD — Path
>
> 85% `edad(_?(actual|a[nñ]os|anios|paciente|afiliado|cliente|al_?(ingreso|diagn[oó]stico|fallecimiento)))?` `grupo_?etario` `rango_?(de_?)?edad` `quinquenio` `age(_?(years|group))?`

> DATA TYPE PE_L2_EDAD — Type
>
> Accepts number (length up to 6) or text (length up to 6); any other type rules the data class out.

HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30–39). A hundred or more becomes 90: rare ages identify.

`PE_EDAD` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

> `PE_CM_DIGITOS` CHARACTER MAPPING
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
37	→	**35**
104	→	**90**

9.4 Personal event dates

PE_L2_FECHA_EVENTO

COLUMN NAME | DATA TYPE | DATE SHIFT

WHAT IT IS

Dates that say something about the person: death, marriage, hiring and leaving, retirement, admission, diagnosis, arrest, sentence, sick leave, DNI issue.

HOW IT IS FOUND

COLUMN NAME PE_L2_FECHA_EVENTO – Path

80% `(fecha|fec|fch|f)_?(de_?)?(defunci[oó]n|fallecimiento|muerte|matrimonio|divorcio|ingreso|cese|egreso|alta|baja|retiro|contrataci[oó]n|despido|jubilaci[oó]n|hospitalizaci[oó]n|internamiento|diagn[oó]stico|atenci[oó]n|consulta|parto|captura|detenci[oó]n|sentencia|condena|emisi[oó]n(_?(dni|documento))?|caducidad_?dni|vacunaci[oó]n|cirug[ií]a|accidente|descanso_?m[eé]dico)` `fec_?(emi|emision|defuncion|ingreso|cese|baja)` `date_?of_?(death|marriage|hire|admission|discharge)`

DATA TYPE PE_L2_FECHA_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted up to 30 days either way.

`PE_FECHA_EVENTO` DATE SHIFT

Moves the date by -30 to 30 days.

EXAMPLES

VALUE		MASKED
2021-03-15	→	**2021-03-05T00:00**

9.5 Sex

PE_L2_SEXO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Sex as the system records it: `F` / `M`, `Femenino` / `Masculino`, `Mujer` / `Hombre` or the codes 1 and 2.

HOW IT IS FOUND

COLUMN NAME PE_L2_SEXO – Path

85% `sexo(_?(biol[oó]gico|al_?nacer|paciente|afiliado|cliente))?` `(tp|tipo|cod|cd|id)_?sexo` `g[eé]nero` *(except when followed by* ***`identidad`****,* ***`autopercibido`****)* `sex` `gender` *(except when followed by* ***`identity`****)*

KNOWN VALUES PE_L2_SEXO – List

LIST	VALUES	STRENGTH	FIRST VALUES
`pe-detectar-sexo.txt`	6	80%	femenino, masculino, mujer, hombre, female, male
`pe-detectar-sexo-letra.txt`	3	60%	f, m, h

Values are split into words, and each word is looked up.

A value that matches none counts 50% against the data class.

HOW IT IS MASKED

Replaced within its own vocabulary: `F` stays `F` or `M`, `1` stays `1` or `2`.

`PE_SEXO` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_SEXO_FEMENINO_MASCULINO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-femenino-masculino.txt`, chosen by a keyed hash.

`PE_SEXO_MUJER_HOMBRE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-mujer-hombre.txt`, chosen by a keyed hash.

`PE_SEXO_FEMALE_MALE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-female-male.txt`, chosen by a keyed hash.

`PE_SEXO_F_M` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-f-m.txt`, chosen by a keyed hash.

`PE_SEXO_H_M` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-h-m.txt`, chosen by a keyed hash.

`PE_SEXO_1_2` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-1-2.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F	→	**M**
Femenino	→	**Femenino**

9.6 Distrito and provincia

PE_L2_DISTRITO

COLUMN NAME VALUE FORMAT DATA CLEANSING

WHAT IT IS

The distrito or provincia of residence, birth or care, by name — alone or with the provincia or the departamento, `Balsas – Amazonas`.

HOW IT IS FOUND

COLUMN NAME PE_L2_DISTRITO – Path

85% `distrito(_?(residencia|nacimiento|domicilio))?` *(except when followed by list E8)* `dist` *(except when followed by* `cod`, `codigo`, `id`*)* `provincia(_?(residencia|nacimiento|domicilio))?` *(except when followed by* `cod`, `codigo`, `id`*)* `prov` *(except when followed by* `cod`, `codigo`, `id`*)* `ciudad(_?(residencia|nacimiento|domicilio|cliente))?` `(nom|nombre|desc)_?(distrito|provincia|ciudad)` `lugar_?(de_?)?nacimiento` `city` `town`

VALUE FORMAT PE_L2_DISTRITO – Values

PATTERN	STRENGTH	NOTES
One of 1,737 known values, ignoring case — Chachapoyas, Asunción, Balsas, Cheto, Chiliquín, Chuquibamba, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A distrito with fewer than 20,000 inhabitants becomes the nearest distrito of its provincia with at least 20,000 — or of its departamento when the provincia has none. The thirty-three provincias under 20,000 follow the same rule inside their departamento. The 25 departamentos stay.

`PE_DISTRITO` DATA CLEANSING

Replaces each value found in `pe-distritos-generalizados.txt` (44,952 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
Asunción	→	**Los Baños del Inca**
Lima	→	**Lima**
Balsas – Amazonas	→	**Chachapoyas – Amazonas**

9.7 UBIGEO

PE_L2_UBIGEO

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The INEI code: two digits for the departamento, four for the provincia and six for the distrito.

HOW IT IS FOUND

COLUMN NAME PE_L2_UBIGEO – Path

85% `ubigeo[a-z0-9_]*` `(cod|codigo|cd|id)_?(ubigeo|distrito|dist|provincia|prov|ciudad|lugar)(_?(inei|reniec|res|residencia|nac|nacimiento))?` `(cod|codigo)_?(inei|reniec|geogr[aá]fico)` `coddist` `codprov`

VALUE FORMAT PE_L2_UBIGEO – Regex

PATTERN	STRENGTH	NOTES
`0*(?:(0[1-9]|1\d|2[0-5])\d{2}(\d{2})?)`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The same generalization as the name: a small distrito's UBIGEO becomes the UBIGEO of the distrito that replaces its name. The code of a large one does not change.

`PE_UBIGEO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `PE_UBIGEO_TABLA` DATA CLEANSING
>
> Replaces each value found in `pe-ubigeo-generalizado.txt` (2,533 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
010102	→	**010101**
150101	→	**150101**

9.8 Centro poblado and caserío

PE_L2_CENTRO_POBLADO

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The centro poblado, caserío, anexo or peasant or native community of a rural address: a few hundred people each, and the address of most rural households.

HOW IT IS FOUND

COLUMN NAME PE_L2_CENTRO_POBLADO — Path

85% `centro_?poblado` `ccpp` `caser[ií]o` `anexo(_?rural)?` `comunidad_?(campesina|nativa|rural)` `comunidad` `aldea` `paraje` `zona_?rural` `sector_?rural` `hacienda` `fundo` `parcela(ci[oó]n)?` `predio_?rural`

HOW IT IS MASKED

Replaced by `Zona rural`.

`PE_ZONA_RURAL` SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-zona-rural.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Caserío El Milagro	→	**Zona rural**

9.9 Urbanización and asentamiento humano

PE_L2_URBANIZACION

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The urbanización, asentamiento humano, pueblo joven, housing cooperative or residencial of residence.

HOW IT IS FOUND

COLUMN NAME PE_L2_URBANIZACION — Path

80% `urbanizaci[oó]n` `urb` `asentamiento_?humano` `aa_?hh` `pueblo_?joven` `asociaci[oó]n_?(de_?)?vivienda` `cooperativa_?(de_?)?vivienda` `conjunto_?habitacional` `residencial` `barrio(_?(residencia|domicilio))?` `(nom|nombre)_?(urbanizacion|barrio|residencial)` `neighbou?rhood`

HOW IT IS MASKED

Replaced by a common Peruvian one, unrelated to the original.

`PE_URBANIZACION` SECURE LOOKUP

Replaces the value with one of the 104 values in `pe-urbanizaciones.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Urb. Santa Patricia	→	**AA.HH. Villa El Salvador**

9.10 Postal code

PE_L2_CODIGO_POSTAL

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The five-digit postal code in use since 2011: the first two are the departamento.

HOW IT IS FOUND

COLUMN NAME PE_L2_CODIGO_POSTAL — Path

85% `c[oó]digo_?postal` `cod_?postal` `codpostal` `zip(_?code)?` `postal_?code` `cp`

DATA TYPE PE_L2_CODIGO_POSTAL — Type

Accepts text (length 4 to 5) or number (length up to 5); any other type rules the data class out.

HOW IT IS MASKED

The first two digits stay and the others become zeros: `15001` becomes `15000`.

`PE_CODIGO_POSTAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
15001	→	**15000**

9.11 Marital status

PE_L2_ESTADO_CIVIL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Single, married, conviviente — the union of fact the Civil Code recognizes —, separated, divorced or widowed, in words or as a code.

HOW IT IS FOUND

COLUMN NAME PE_L2_ESTADO_CIVIL — Path

85% `estado_?civil` `est_?civil` `(cod|tipo|id)_?estado_?civil` `situaci[oó]n_?conyugal` `uni[oó]n_?de_?hecho` `marital_?status` `civil_?status`

VALUE FORMAT PE_L2_ESTADO_CIVIL — Values

PATTERN	STRENGTH	NOTES
One of 20 known values, ignoring case — Soltero a, Casado a, Conviviente, Separado a, Divorciado a, Viudo a, ...	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another status from the list; a code stays a code.

`PE_ESTADO_CIVIL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ESTADO_CIVIL_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

> Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_ESTADO_CIVIL_CODIGO` SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-estado-civil-codigos.txt`, chosen by a keyed hash.

`PE_ESTADO_CIVIL_LISTA` SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-estado-civil.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Unión de hecho	→	**Soltero/a**

9.12 Occupation

PE_L2_OCUPACION

COLUMN NAME · REGEX DECOMPOSE

WHAT IT IS

Occupation, profession or trade, in words or by the national occupation classification code. A rare occupation in a small distrito identifies on its own.

HOW IT IS FOUND

> COLUMN NAME — PE_L2_OCUPACION — Path
>
> **80%** `ocupaci[oó]n` `profesi[oó]n` `oficio` `cno` `(cod|codigo)_?(ocupacion|profesion|cno)` `puesto_?(de_?)?trabajo` `categor[ií]a_?ocupacional` `cargo_?(actual|empleado)` `occupation` `profession` `job_?title`
>
> **50%** `cargo` `puesto`

HOW IT IS MASKED

Text is replaced by a common occupation; a code is mapped digit by digit.

`PE_OCUPACION_O_CODIGO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_OCUPACION`.

> `PE_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `PE_OCUPACION` SECURE LOOKUP
>
> Replaces the value with one of the 39 values in `pe-ocupaciones.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Gerente de operaciones	→	**Programador**
2221	→	**1980**

9.13 Employer

PE_L2_EMPLEADOR

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The name of the company or institution where the person works.

HOW IT IS FOUND

COLUMN NAME PE_L2_EMPLEADOR — Path

90% `empleador(_?(nombre|raz[oó]n_?social|ruc))?` `(nombre|nom)_?(empleador|empresa_?donde_?trabaja|patrono)` `empresa_?(donde_?)?(trabaja|labora)` `lugar_?(de_?)?trabajo` `centro_?(de_?)?(trabajo|labores)` `entidad_?empleadora` `employer(_?name)?` `workplace`

HOW IT IS MASKED

Replaced by a fictitious company.

`PE_EMPLEADOR` SECURE LOOKUP

Replaces the value with one of the 24 values in `pe-empleadores.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Corporación Comercial Peruana S.A.C.	→	**Panadería Pan de Casa**

9.14 Education

PE_L2_NIVEL_EDUCATIVO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The education level attained — from none to a doctorate — in the categories the census uses, in words or as a code.

HOW IT IS FOUND

COLUMN NAME PE_L2_NIVEL_EDUCATIVO — Path

80% `escolaridad` `nivel_?(educativo|de_?estudios|acad[eé]mico|alcanzado|de_?instrucci[oó]n)` `grado_?(de_?)?(estudio|instrucci[oó]n|escolaridad)` `m[aá]ximo_?nivel` `a[nñ]os_?(de_?)?estudio` `education(_?level)?`

VALUE FORMAT PE_L2_NIVEL_EDUCATIVO — Values

PATTERN	STRENGTH	NOTES
One of 21 known values, ignoring case — Sin nivel, Inicial, Primaria incompleta, Primaria completa, Secundaria incompleta, Secundaria completa, …	60%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another level from the list; a code stays a code.

`PE_NIVEL_EDUCATIVO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_NIVEL_EDUCATIVO_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_NIVEL_EDUCATIVO_CODIGO` SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo-codigos.txt`, chosen by a keyed hash.

`PE_NIVEL_EDUCATIVO_LISTA` SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Educación superior en curso	→	**Maestría o doctorado**

9.15 School

PE_L2_INSTITUCION_EDUCATIVA

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The institución educativa, CETPRO, institute or university, by name or by the Ministry of Education modular code.

HOW IT IS FOUND

COLUMN NAME PE_L2_INSTITUCION_EDUCATIVA — Path

75% `instituci[oó]n_?educativa` `i_?e(_?p)?` `centro_?(educativo|de_?estudios)` `cetpro` `(nombre|nom)_?(institucion_?educativa|escuela|colegio|universidad)` `escuela` `colegio` `instituto` `universidad` `c[oó]digo_?modular` `c[oó]digo_?(de_?)?local_?escolar` `school(_?name)?` `university`

HOW IT IS MASKED

Replaced by a fictitious school.

`PE_INSTITUCION_EDUCATIVA` SECURE LOOKUP

Replaces the value with one of the 12 values in `pe-instituciones-educativas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Universidad Nacional de Ingeniería	→	**Instituto Superior Tecnológico del Norte**

9.16 Dependants and children

PE_L2_PERSONAS_A_CARGO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Number of dependants, children, derechohabientes or household members.

HOW IT IS FOUND

COLUMN NAME PE_L2_PERSONAS_A_CARGO — Path

80% `personas_?a_?cargo` `n(o|ro|um|umero)_?(de_?)?(hijos|dependientes|derechohabientes|personas_?a_?cargo)` `cant(idad)?_?(hijos|dependientes|personas_?hogar|miembros)` `hijos` `dependientes` `derechohabientes` `personas_?(en_?el_?)?hogar` `tama[nñ]o_?(del_?)?hogar` `miembros_?(del_?)?hogar` `number_?of_?children` `dependents`

DATA TYPE PE_L2_PERSONAS_A_CARGO — Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large households are rare, and a rare count identifies.

`PE_CONTEO_LIMITADO` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
9	→	**5**

PART 10

Sensitive data (art. 2, 5)

The list of article 2, numeral 5 is short and blunt: biometric data that identifies on its own, racial and ethnic origin, economic income, political, religious, philosophical and moral opinions or convictions, trade union membership, and information on health or sexual life. Income lives in the financial group, where the money is, and health in a group of its own. Here are the rest, together with the genetic data, the gender identity, the indigenous people and language, the nationality and the victim status that other Peruvian statutes protect and that re-identify just as fast.

10.1 Ethnic origin

PE_L3_ORIGEN_ETNICO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Self-identification as quechua, aimara, native of the Amazon, afroperuano, white, mestizo, nikkei or tusán — the categories the 2017 census asked everyone from twelve up.

HOW IT IS FOUND

COLUMN NAME — PE_L3_ORIGEN_ETNICO — Path

95% `etnia` `origen_?([eé]tnico|racial)` `grupo_?[eé]tnico` `pertenencia_?[eé]tnica` `autoidentificaci[oó]n(_?[eé]tnica)?` `autorreconocimiento` `(cod|codigo|tipo|id)_?(etnia|origen_?etnico)` `raza` `ind[ií]gena` `afroperuano` `afrodescendiente` `ethnicity` `race`

VALUE FORMAT — PE_L3_ORIGEN_ETNICO — Values

PATTERN	STRENGTH	NOTES
One of 23 known values, ignoring case — Quechua, Aimara, Nativo o indígena de la Amazonía, Perteneciente a otro pueblo indígena u originario, Negro moreno zambo mulato o afroperuano, Blanco, …	90%	
One of 10 known values, ignoring case — s, n, sí, no, 1, 2, …	30%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a category from the list.

`PE_ORIGEN_ETNICO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ORIGEN_ETNICO_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_ORIGEN_ETNICO_CODIGO` SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-origen-etnico-codigos.txt`, chosen by a keyed hash.

`PE_ORIGEN_ETNICO_LISTA` SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-origen-etnico.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Indígena	→	**Perteneciente a otro pueblo indígena u originario**
S	→	**S**

10.2 Indigenous people

PE_L3_PUEBLO_INDIGENA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The people of the Base de Datos Oficial de Pueblos Indígenas u Originarios the person belongs to, and the peasant or native community. Naming the people is naming the ethnic origin.

HOW IT IS FOUND

COLUMN NAME PE_L3_PUEBLO_INDIGENA — Path

90% `pueblo_?(ind[ií]gena|originario)` `comunidad_?(nativa|campesina)_?(nombre|pertenencia)` `pertenece_?(a_?)?(pueblo|comunidad)` `bdpi` `base_?de_?datos_?(de_?)?pueblos` `etnia_?` `amaz[oó]nica` `indigenous_?people`

VALUE FORMAT PE_L3_PUEBLO_INDIGENA — Values

PATTERN	STRENGTH	NOTES
One of 41 known values, ignoring case — Quechuas, Aimara, Asháninka, Awajún, Shipibo-Konibo, Shawi, …	85%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another people from the list; a flag stays a flag.

`PE_PUEBLO_INDIGENA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_PUEBLO_INDIGENA_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_PUEBLO_INDIGENA_LISTA` SECURE LOOKUP

Replaces the value with one of the 21 values in `pe-pueblos-indigenas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Pueblo Ese Eja	→	**Ticuna**

10.3 Language

PE_L3_LENGUA

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The mother or indigenous language. Peru has forty-eight beside Spanish, and the Ley 29735 makes them official where they prevail: the language says where a person comes from.

HOW IT IS FOUND

COLUMN NAME — PE_L3_LENGUA — Path

90% `lengua(_?(ind[ií]gena|materna|nativa|originaria|de_?se[ñn]as))?` `idioma(_?(ind[ií]gena|materno|nativo|originario))?` `habla_?(lengua|idioma)` `lengua_?que_?aprendi[oó]` `mother_?tongue` `native_?language`

VALUE FORMAT — PE_L3_LENGUA — Values

PATTERN	STRENGTH	NOTES
One of 28 known values, ignoring case — Castellano, Quechua, Aimara, Asháninka, Awajún, Shipibo-Konibo, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a language from the list; a flag stays a flag.

`PE_LENGUA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_LENGUA_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_LENGUA_LISTA` SECURE LOOKUP

Replaces the value with one of the 18 values in `pe-lenguas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Quechua chanka	→	**Shipibo-Konibo**

10.4 Nationality

PE_L3_NACIONALIDAD

COLUMN NAME · KNOWN VALUES · SECURE LOOKUP

WHAT IT IS

Nationality, country of origin and migratory status. The law does not list it, but it sits beside the racial and ethnic origin it names, and in Peru it separates the nationals from the Venezuelan community.

HOW IT IS FOUND

COLUMN NAME PE_L3_NACIONALIDAD — Path

85% `nacionalidad` `pa[ií]s_?(de_?)?(nacimiento|origen|nacionalidad|procedencia)` `ciudadan[ií]a` `nationality` `citizenship` `country_?of_?birth` `condici[oó]n_?(de_?)?(migratoria|refugiado)` `persona_?refugiada` `migrante`

KNOWN VALUES PE_L3_NACIONALIDAD — List

LIST	VALUES	STRENGTH	FIRST VALUES
`pe-detectar-nacionalidades.txt`	41	70%	Peruana, Venezolana, Colombiana, Ecuatoriana, Boliviana, Chilena

Values are split into words, and each word is looked up.

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a nationality from the list: the Peruvian one and the largest foreign communities.

`PE_NACIONALIDAD` SECURE LOOKUP

Replaces the value with one of the 14 values in `pe-nacionalidades.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Panameña	→	**Japonesa**

10.5 Religion

PE_L3_RELIGION

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The religion, church or congregation the person belongs to.

HOW IT IS FOUND

COLUMN NAME PE_L3_RELIGION — Path

95% `religi[oó]n` `creencia_?(religiosa|espiritual)` `convicci[oó]n_?(religiosa|espiritual)` `credo` `confesi[oó]n_?religiosa` `culto` `iglesia(_?(a_?la_?que_?pertenece|nombre))?` `denominaci[oó]n_?religiosa` `(cod|codigo|tipo)_?religion` `religion` `church`

VALUE FORMAT PE_L3_RELIGION — Values

PATTERN	STRENGTH	NOTES
One of 26 known values, ignoring case — Católica, Evangélica, Cristiana, Adventista, Testigo de Jehová, Iglesia de Jesucristo de los Santos de los Últimos Días, …	90%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a religion from the list.

`PE_RELIGION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_RELIGION_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_RELIGION_LISTA` SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-religiones.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Ortodoxa	→	**Cristiana**

10.6 Philosophical and moral convictions

PE_L3_CONVICCION_FILOSOFICA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The philosophical and moral convictions article 2, 5 names beside the religious ones, conscientious objection and belonging to a lodge.

HOW IT IS FOUND

COLUMN NAME PE_L3_CONVICCION_FILOSOFICA — Path

90% `convicci[oó]n(es)?_?(filos[oó]ficas?|morales?|espirituales?|personales?)` `creencias?_?(filos[oó]ficas?|morales?|personales?)` `opini[oó]n_?(filos[oó]fica|moral)` `objeci[oó]n_?(de_?)?conciencia` `objetor_?(de_?)?conciencia` `logia` `masoner[ií]a` `beliefs?` `philosophical_?beliefs?`

HOW IT IS MASKED

The values share no vocabulary: they are suppressed to `Sin información`. Flags and codes keep their shape.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Se niega a transfusiones	→	**Sin información**

10.7 Political opinion

PE_L3_OPINION_POLITICA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The political opinion article 2, 5 names: leaning, sympathy and voting intention, from surveys and records.

HOW IT IS FOUND

COLUMN NAME PE_L3_OPINION_POLITICA — Path

90% `ideolog[ií]a(_?pol[ií]tica)?` `opini[oó]n_?pol[ií]tica` `orientaci[oó]n_?pol[ií]tica` `tendencia_?pol[ií]tica` `preferencia_?(pol[ií]tica|partidaria|electoral)` `intenci[oó]n_?(de_?)?voto` `simpat[ií]a_?pol[ií]tica` `political_?(opinion|orientation|view)` `voting_?intention`

VALUE FORMAT PE_L3_OPINION_POLITICA — Values

PATTERN	STRENGTH	NOTES
One of 16 known values, ignoring case — Izquierda, Centroizquierda, Centro, Centroderecha, Derecha, Independiente, …	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a position from the list — left to right, independent.

`PE_IDEOLOGIA_POLITICA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_IDEOLOGIA_POLITICA_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_IDEOLOGIA_POLITICA_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `pe-ideologias-politicas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Oficialista`	→	**`Centro`**

10.8 Party membership

PE_L3_AFILIACION_PARTIDARIA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The political party the person belongs to, from the Registro de Organizaciones Políticas of the JNE.

HOW IT IS FOUND

COLUMN NAME PE_L3_AFILIACION_PARTIDARIA — Path

95% `partido(_?pol[ií]tico)?(_?(afiliaci[oó]n|nombre))?` `organizaci[oó]n_?pol[ií]tica` `afiliaci[oó]n_?(pol[ií]tica|partidaria|partido)` `afiliado_?partido` `militancia(_?pol[ií]tica)?` `militante` `(cod|codigo|nombre|nom)_?partido` `political_?party` `party_?membership`

VALUE FORMAT PE_L3_AFILIACION_PARTIDARIA — Values

PATTERN	STRENGTH	NOTES
One of 18 known values, ignoring case — Fuerza Popular, Perú Libre, Acción Popular, Alianza para el Progreso, Renovación Popular, Avanza País, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another party from the list; a flag stays a flag.

`PE_PARTIDO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_PARTIDO_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_PARTIDO_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 13 values in `pe-partidos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Frente Amplio	→	**Perú Libre**
S	→	**S**

10.9 Trade union membership

PE_L3_AFILIACION_SINDICAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The union, membership, the union dues deducted on the payroll and union protection. Article 2, 5 names it outright.

HOW IT IS FOUND

COLUMN NAME PE_L3_AFILIACION_SINDICAL — Path

95% `sindicato(_?(nombre|afiliado|afiliaci[oó]n|codigo))?` `sindicalizado` `afiliado_?sindicato` `afiliaci[oó]n_?sindical` `cuota_?sindical` `descuento_?sindical` `aporte_?sindical` `asociaci[oó]n_?sindical` `convenio_?colectivo` `fuero_?sindical` `dirigente_?sindical` `trade_?union` `union_?member(ship)?`

VALUE FORMAT PE_L3_AFILIACION_SINDICAL — Values

PATTERN	STRENGTH	NOTES
One of 11 known values, ignoring case — CGTP, CUT Perú, CTP, CATP, SUTEP, FENTAP, …	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
SUTEP	→	**`Sin información`**
S	→	**`S`**

10.10 Biometric data

PE_L3_BIOMETRICO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Fingerprint templates — the DNI's fingerprint is the country's identification pattern —, face, iris, voice and digitized signature. Article 2, 5 makes sensitive the biometric data that **by itself** can identify the subject.

HOW IT IS FOUND

COLUMN NAME PE_L3_BIOMETRICO — Path

90% `biometr[ií]a[a-z_]*` `biom[eé]trico[a-z_]*` `huella(_?(dactilar|digital))?(_?(template|plantilla|imagen|hash))?` `huellas` `dactilar` `template_?(facial|huella|biometrico)` `plantilla_?(facial|biom[eé]trica)` `rostro_?(hash|template)` `reconocimiento_?facial` `firma_?(biom[eé]trica|digitalizada|electr[oó]nica)` `iris` `voz_?(biom[eé]trica|template)` `fingerprints?` `face_?(template|hash|encoding)` `biometric[a-z_]*`

DATA TYPE PE_L3_BIOMETRICO — Type

Accepts text; any other type rules the data class out.

HOW IT IS MASKED

Every character becomes `X`. A masked template authenticates nobody.

`PE_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
A1B2C3D4E5F6	→	XXXXXXXXXXXX

10.11 Genetic data

PE_L3_GENETICO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Results of genetic and DNA tests, paternity tests, genetic profiles, mutations, karyotype and newborn screening.

HOW IT IS FOUND

COLUMN NAME PE_L3_GENETICO — Path

90% `adn` `dna` `gen[eé]tic[oa]s?` `datos?_?gen[eé]ticos?` `genoma` `genotip[a-z_]*` `prueba_?(gen[eé]tica|de_?adn|de_?paternidad|de_?filiaci[oó]n)` `perfil_?gen[eé]tico` `mutaci[oó]n` `brca[12]?` `cariotipo` `tamizaje_?neonatal` `genetic[a-z_]*`

VALUE FORMAT PE_L3_GENETICO — Regex

PATTERN	STRENGTH	NOTES
`[ACGTUN]{24,}`	90%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`.

`PE_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
BRCA1 positivo	→	XXXXXXXXXXXXXX

10.12 Sexual orientation

PE_L3_ORIENTACION_SEXUAL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Self-declared sexual orientation or preference, in health and diversity records. It belongs to the *vida sexual* article 2, 5 protects.

HOW IT IS FOUND

COLUMN NAME PE_L3_ORIENTACION_SEXUAL — Path

95% `orientaci[oó]n_?sexual` `preferencias?_?sexuales?` `(cod|codigo|tipo)_?orientacion_?sexual` `sexual_?orientation`

VALUE FORMAT PE_L3_ORIENTACION_SEXUAL — Values

PATTERN	STRENGTH	NOTES
One of 13 known values, ignoring case — Heterosexual, Gay, Lesbiana, Bisexual, Pansexual, Asexual, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a category from the list.

`PE_ORIENTACION_SEXUAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ORIENTACION_SEXUAL_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_ORIENTACION_SEXUAL_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `pe-orientaciones-sexuales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Homosexual`	→	**`Pansexual`**

10.13 Sexual life

PE_L3_VIDA_SEXUAL

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Sexual activity and behaviour, number of partners, age at first intercourse — named word for word in article 2, 5.

HOW IT IS FOUND

> COLUMN NAME PE_L3_VIDA_SEXUAL – Path
>
> 90% `vida_?sexual` `actividad_?sexual` `conducta_?sexual` `pr[aá]cticas?_?sexuales?` `parejas?_?sexuales?` `n(o|ro|um|umero)_?parejas` `sexualmente_?activ[oa]` `inicio_?(de_?)?(la_?)?vida_?sexual` `relaciones_?sexuales` `sex(ual)?_?life` `sexual_?(activity|behavio(u)?r|partners)`

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Activa	→	**Sin información**

10.14 Gender identity

PE_L3_IDENTIDAD_GENERO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Self-declared gender identity and the social name. The law does not name it, but it travels with the sexual life it does name.

HOW IT IS FOUND

COLUMN NAME PE_L3_IDENTIDAD_GENERO — Path

95% `identidad_?(de_?)?g[eé]nero` `(cod|codigo|tipo)_?identidad_?genero` `g[eé]nero_?(autopercibido|identitario)` `expresi[oó]n_?de_?g[eé]nero` `transg[eé]nero` `persona_?trans` `nombre_?social` `pronombres?` `gender_?identity`

VALUE FORMAT PE_L3_IDENTIDAD_GENERO — Values

PATTERN	STRENGTH	NOTES
One of 15 known values, ignoring case — Mujer cisgénero, Hombre cisgénero, Mujer trans, Hombre trans, Persona no binaria, Otra, …	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a category from the list.

`PE_IDENTIDAD_GENERO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_IDENTIDAD_GENERO_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_IDENTIDAD_GENERO_LISTA` SECURE LOOKUP

Replaces the value with one of the 7 values in `pe-identidades-genero.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Transgénero	→	**Hombre trans**

10.15 Victims

PE_L3_VICTIMA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The Registro Único de Víctimas of the Ley 28592 records the people harmed by the violence of 1980 to 2000; the Ley 30364 protects the identity of the victims of violence against women and members of the family group.

HOW IT IS FOUND

COLUMN NAME PE_L3_VICTIMA – Path

90% `v[ií]ctima(_?(de_?)?(violencia(_?(de_?g[eé]nero|familiar|sexual|econ[oó]mica|psicol[oó]gica))?|delito|trata|desplazamiento|terrorismo|abuso))?` `ruv` `registro_?[uú]nico_?(de_?)?v[ií]ctimas` `violencia_?(de_?)?g[eé]nero` `violencia_?(familiar|dom[eé]stica|sexual|psicol[oó]gica)` `tipo_?(de_?)?violencia` `medida_?(de_?)?protecci[oó]n` `ficha_?de_?valoraci[oó]n_?(de_?)?riesgo` `desplazamiento_?forzado` `trata_?(de_?)?personas` `denunciante_?violencia` `cem`

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

> `PE_BANDERA` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_SUPRIMIR` **SECURE LOOKUP**
>
> Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Violencia familiar	→	**Sin información**
S	→	**S**

PART 11

Health data

Article 2, 5 makes *information related to health* sensitive without qualification, and the Ley 26842 General de Salud adds medical secrecy on top. The group keeps the clinical record apart from the rest so that a health database can be masked whole.

11.1 Health identifiers

PE_SALUD_IDENTIFICADOR

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Clinical record, admission, care, appointment, prescription and FUA numbers, and the RENHICE identifier of the electronic record.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_IDENTIFICADOR – Path

85% `n(o|ro|um|umero)_?(historia(_?cl[ií]nica)?|hc|hce|expediente(_?cl[ií]nico)?|ingreso|atenci[oó]n|consulta|cita|referencia_?m[eé]dica|receta|fua)` `historia_?cl[ií]nica` `hc_?(n(o|ro|um|umero))?` `hce` `renhice` `n(o|ro|um|umero)_?(asegurado|afiliado_?(sis|essalud))` `c[oó]digo_?(paciente|historia|atenci[oó]n)` `id_?paciente`

55% `expediente` `atenci[oó]n` `ingreso` `consulta`

HOW IT IS MASKED

Character by character, keeping length and format.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
HC–2024–00123	→	**TW–3389–01317**

11.2 Health insurance

PE_SALUD_COBERTURA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

EsSalud, the SIS, an EPS, the health service of the armed forces or the police, or a private insurer, by name or code. Which one a person has says a great deal about them.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_COBERTURA – Path

85% `cobertura(_?(m[eé]dica|de_?salud|social))?` `seguro_?(m[eé]dico|salud|integral)` `r[eé]gimen(_?(de_?)?salud)?` `tipo_?(de_?)?seguro` `proveedor_?(de_?)?salud` `essalud(_?(afiliado|cobertura))?` `sis(_?(afiliado|tipo))?` `eps` `iafas` `sanidad_?(pnp|ffaa|militar|policial)` `aseguradora_?(m[eé]dica|salud)` `plan_?(m[eé]dico|de_?salud)`

VALUE FORMAT PE_SALUD_COBERTURA — Values

PATTERN	STRENGTH	NOTES
One of 15 known values, ignoring case — EsSalud, SIS Seguro Integral de Salud, EPS, Sanidad de las Fuerzas Armadas, Sanidad de la Policía Nacional, Seguro privado de salud, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A name is replaced by another cover from the list; a code is mapped digit by digit; a flag stays a flag.

`PE_COBERTURA_SALUD` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_COBERTURA` .

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.
>
> `PE_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `PE_COBERTURA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `pe-cobertura-salud.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
EsSalud	→	**Seguro privado de salud**
12345	→	**31564**

11.3 Diagnosis and ICD-10

PE_SALUD_DIAGNOSTICO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Diagnoses in words or by ICD-10 code, with a dot (`F33.1`) or without (`F331`); causes of death, of sick leave and of admission; allergies and comorbidities.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_DIAGNOSTICO — Path

85% `diagn[oó]stico[a-z_]*` `dx(_?(principal|secundario|ingreso|egreso)[0-9]?)?` `cie_?10` `cie(_?(10|11|principal))?` `(cod|codigo)_?(diagnostico|dx|cie)` `impresi[oó]n_?diagn[oó]stica` `enfermedad(es)?[a-z_]*` `patolog[ií]a` `comorbilidad(es)?` `antecedentes?_?(m[eé]dicos|patol[oó]gicos|cl[ií]nicos|personales)` `causa_?(de_?)?(muerte|defunci[oó]n|incapacidad|hospitalizaci[oó]n|consulta)` `motivo_?(de_?)?(consulta|descanso|hospitalizaci[oó]n)` `alergias?` `diagnos(is|es)` `icd_?(10|11)?`

VALUE FORMAT PE_SALUD_DIAGNOSTICO — Regex

PATTERN	STRENGTH	NOTES
`[A-TV-Za-tv-z]\d{2}(\.\d{1,2}|\d|X)?`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

An ICD-10 code becomes another common code in the same format, and text a common diagnosis: an HIV or mental disorder diagnosis does not survive as such.

`PE_DIAGNOSTICO` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PE_DIAGNOSTICO_TEXTO`.

> `PE_CIE10_DECIMAL` **SECURE LOOKUP**
>
> Replaces the value with one of the 33 values in `pe-cie10-decimal.txt`, chosen by a keyed hash.
>
> `PE_CIE10` **SECURE LOOKUP**
>
> Replaces the value with one of the 37 values in `pe-cie10.txt`, chosen by a keyed hash.
>
> `PE_BANDERA` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_DIAGNOSTICO_TEXTO` **SECURE LOOKUP**
>
> Replaces the value with one of the 30 values in `pe-diagnosticos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F33.1	→	**K21.9**
B24X	→	**R51**

11.4 Procedures

PE_SALUD_PROCEDIMIENTO

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Procedures, tests and surgeries, in words or by code. A viral load or a pregnancy test reveals the diagnosis.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_PROCEDIMIENTO — Path

85% `procedimiento(_?(m[eé]dico|quir[uú]rgico|realizado|autorizado))?` `(cod|codigo)_?(procedimiento|prestaci[oó]n|servicio_?salud|cpt)` `cpt` `prestaci[oó]n_?(m[eé]dica|de_?salud)` `examen_?(ordenado|realizado)` `cirug[ií]a(_?realizada)?` `intervenci[oó]n_?quir[uú]rgica` `medical_?procedure`

HOW IT IS MASKED

A code is mapped digit by digit; text is replaced by a common procedure.

`PE_PROCEDIMIENTO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_PROCEDIMIENTO_TEXTO`.

> `PE_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `PE_PROCEDIMIENTO_TEXTO` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `pe-procedimientos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
901234	→	**928120**

11.5 Medicines

PE_SALUD_MEDICAMENTO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Prescribed or dispensed medicines, by name or active ingredient. An antiretroviral, an antipsychotic or a contraceptive says what the diagnosis column was masked for.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_MEDICAMENTO — Path

85% `medicamentos?(_?(recetado|prescrito|entregado|despachado|nombre|uso))?` `f[aá]rmacos?` `principio_?activo` `denominaci[oó]n_?com[uú]n_?internacional` `dci` `receta(_?m[eé]dica)?` `prescripci[oó]n(_?medicamento)?` `posolog[ií]a` `tratamiento_?farmacol[oó]gico` `(cod|codigo)_?medicamento` `petitorio_?(nacional_?)?[uú]nico` `medications?` `drugs?_?prescribed`

VALUE FORMAT PE_SALUD_MEDICAMENTO — Values

PATTERN	STRENGTH	NOTES
One of 46 known values, ignoring case — Paracetamol, Ibuprofeno, Diclofenaco, Metformina, Glibenclamida, Losartán, …	70%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced by a common medicine from the list; a flag stays a flag.

`PE_MEDICAMENTO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_MEDICAMENTO_LISTA` .

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt` , chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_MEDICAMENTO_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 22 values in `pe-medicamentos.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Sertralina 50 mg	→	**Complejo B**

11.6 Clinical text

PE_SALUD_TEXTO_CLINICO

COLUMN NAME | DATA TYPE | FREE TEXT REDACTION

WHAT IT IS

History, present illness, progress notes, physical examination, treatment plan, discharge summaries and nursing notes.

HOW IT IS FOUND

> COLUMN NAME PE_SALUD_TEXTO_CLINICO — Path
>
> 85% `anamnesis` `evoluci[oó]n(_?(m[eé]dica|cl[ií]nica|enfermer[ií]a))?` `enfermedad_?actual` `examen_?f[ií]sico` `plan_?(de_?)?(manejo|tratamiento)` `indicaciones_?m[eé]dicas` `conducta(_?m[eé]dica)?` `epicrisis` `resumen_?(de_?)?(alta|egreso|atenci[oó]n)` `nota_?(m[eé]dica|de_?enfermer[ií]a|cl[ií]nica)` `notas?_?(m[eé]dicas|cl[ií]nicas)` `triaje(_?texto)?` `informe_?(patolog[ií]a|radiolog[ií]a|m[eé]dico)` `clinical_?notes?` `discharge_?summary`

> DATA TYPE PE_SALUD_TEXTO_CLINICO — Type
>
> Accepts text (length 30 or more); any other type rules the data class out.

HOW IT IS MASKED

The same Free Text Redaction as free text: documents, contacts and names from the list are redacted. The clinical content stays, because that is what makes the copy useful.

`PE_TEXTO_LIBRE` FREE TEXT REDACTION

Replaces the text matching 8 patterns with `[DATO]`, and the 1,906 terms of `pe-texto-nombres.txt` with `[NOMBRE]`.

EXAMPLES

VALUE		MASKED
`Paciente Quispe, DNI 45678901, consulta por cefalea.`	→	**`Paciente [NOMBRE], DNI [DATO], consulta por cefalea.`**

11.7 Test results

PE_SALUD_RESULTADO_EXAMEN

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Laboratory and test results — glucose, pregnancy, toxicology, blood alcohol —, also as numeric values.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_RESULTADO_EXAMEN – Path

80% `resultado_?(del_?)?` `(examen|prueba|laboratorio|pcr|serolog[ií]a|biopsia|glucosa|citolog[ií]a|papanicolaou)` `(examen|prueba)_?(resultado|laboratorio)` `glucemia` `hemoglobina(_?glicosilada)?` `hba1c` `colesterol` `imc` `presi[oó]n_?arterial` `prueba_?(de_?)?embarazo` `toxicol[oó]gico` `dosaje_?et[ií]lico` `alcoholemia` `lab_?results?` `test_?results?`

50% `resultado` `prueba`

HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`Reactivo`	→	**`Quilqaju`**

11.8 HIV and sexually transmitted infections

PE_SALUD_VIH

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Serological status, viral load and antiretroviral treatment, hepatitis, tuberculosis and sexually transmitted infections. The Ley 26626 orders the result of an HIV test to be confidential and forbids demanding it to get or keep a job.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_VIH — Path

90% `vih(_?(estado|resultado|prueba|diagn[oó]stico|positivo))?` `hiv` `sida` `aids` `serolog[ií]a_?(vih|hiv)` `estado_?serol[oó]gico` `carga_?viral` `cd4` `targa` `tarv` `antirretroviral(es)?` `its` `ets` `infecci[oó]n(es)?_?(de_?)?transmisi[oó]n_?sexual` `s[ií]filis` `hepatitis_?[bc]` `tuberculosis` `tbc`

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Reactivo	→	**Sin información**
S	→	**S**

11.9 Blood group and Rh

PE_SALUD_GRUPO_SANGUINEO

COLUMN NAME VALUE FORMAT SECURE LOOKUP

WHAT IT IS

Blood group and Rh factor.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_GRUPO_SANGUINEO — Path

90% `grupo_?sangu[ií]neo` `tipo_?(de_?)?sangre` `rh` `factor_?rh` `gs_?rh` `grupo_?rh` `abo(_?rh)?` `blood_?(type|group)`

VALUE FORMAT PE_SALUD_GRUPO_SANGUINEO — Values

PATTERN	STRENGTH	NOTES
One of 18 known values, ignoring case — A, A-, B, B-, AB, AB-, …	90%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another group from the list.

`dlpx-core:BloodType SL` SECURE LOOKUP

Comes with Delphix and is used as it ships. Replaces the blood group with one of the eight groups, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
0+	→	**0−**

11.10 Disability

PE_SALUD_DISCAPACIDAD

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Disability status and type, the CONADIS registration and the disability certificate, under the Ley 29973.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_DISCAPACIDAD — Path

90% `discapacidad[a-z_]*` `(tipo|cod|codigo|categoria)_?discapacidad` `persona_?con_?discapacidad` `pcd` `conadis` `carn[eé]_?(de_?)?(conadis|discapacidad)` `certificado_?(de_?)?discapacidad` `condici[oó]n_?(de_?)?discapacidad` `necesidades_?(educativas_?)?especiales` `movilidad_?reducida` `disabilit(y|ies)`

VALUE FORMAT PE_SALUD_DISCAPACIDAD — Values

PATTERN	STRENGTH	NOTES
One of 22 known values, ignoring case — Física o motora, Auditiva, Visual, Sordoceguera, Intelectual, Mental o psicosocial, …	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a category from the list.

`PE_DISCAPACIDAD` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_DISCAPACIDAD_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_DISCAPACIDAD_LISTA` SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-discapacidades.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Trastorno del espectro autista	→	**Ninguna**

11.11 Occupational health

PE_SALUD_OCUPACIONAL

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The fitness opinion of occupational medical exams (EMO), work accidents and occupational diseases, the SCTR, sick leave and absenteeism.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_OCUPACIONAL — Path

85% `aptitud_?(laboral|m[eé]dica)` `concepto_?(de_?)?aptitud` `examen_?(m[eé]dico_?)?(ocupacional|preocupacional|de_?ingreso|de_?retiro|peri[oó]dico)` `emo` `accidente_?(de_?)?trabajo` `enfermedad_?(laboral|profesional|ocupacional)` `riesgo_?(profesional|laboral)` `sctr` `incapacidad(_?(m[eé]dica|laboral|d[ií]as))?` `descanso_?m[eé]dico` `d[ií]as_?(de_?)?(incapacidad|descanso)` `licencia_?(m[eé]dica|por_?enfermedad|de_?maternidad)` `ausentismo` `comit[eé]_?(de_?)?seguridad`

VALUE FORMAT PE_SALUD_OCUPACIONAL — Values

PATTERN	STRENGTH	NOTES
One of 5 known values, ignoring case — apto, no apto, apto con restricciones, pendiente, apto con recomendaciones	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

An opinion is replaced by another from the list — fit, unfit —; a flag stays a flag; the diagnosis falls under its own data class.

`PE_APTITUD_LABORAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_APTITUD_LABORAL_LISTA`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_APTITUD_LABORAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 4 values in `pe-aptitud-laboral.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
No apto temporalmente	→	**Pendiente**

11.12 Sexual and reproductive health

PE_SALUD_REPRODUCTIVA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Pregnancy, last menstrual period, prenatal care, childbirth, family planning, fertility.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_REPRODUCTIVA — Path

90% `embarazo` `embarazada` `gestante` `gestaci[oó]n` `edad_?gestacional` `semanas_?(de_?)?gestaci[oó]n` `fum` `fecha_?[uú]ltima_?(regla|menstruaci[oó]n)` `control_?prenatal` `prenatal` `parto(_?tipo)?` `ces[aá]rea` `aborto` `m[eé]todo_?(de_?)?planificaci[oó]n` `planificaci[oó]n_?familiar` `anticoncepci[oó]n` `anticonceptivo` `fertilidad` `salud_?(sexual|reproductiva)` `pregnan(t|cy)` `contracepti(on|ve)`

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
12 semanas de gestación	→	**Sin información**
S	→	**S**

11.13 Mental health

PE_SALUD_MENTAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Mental disorders, psychiatric or psychological care, suicide attempts and substance use.

HOW IT IS FOUND

COLUMN NAME PE_SALUD_MENTAL — Path

90% `salud_?mental` `trastorno[a-z_]*` `psiqui[aá]tri[a-z_]*` `psicol[oó]gi[a-z_]*` `depresi[oó]n` `ansiedad` `suicid[a-z_]*` `intento_?(de_?)?suicidio` `autolesi[oó]n` `consumo_?(de_?)?(sustancias|alcohol|drogas)` `adicci[oó]n(es)?` `alcoholismo` `tabaquismo` `mental_?health`

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información`.

`PE_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`PE_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Trastorno bipolar	→	**Sin información**

PART 12

Financial and socioeconomic data

Economic income is sensitive data in Peru (art. 2, 5) — no neighbouring law says so. The rest of the group is not sensitive by that article, but it identifies: a debt, an asset, a pension, a social programme or a SISFOH classification narrows a person down as fast as a postcode.

12.1 Income and pay

PE_FIN_INGRESOS

COLUMN NAME · DATA TYPE · NUMERIC MAPPING

WHAT IT IS

Salary, monthly or household income, fees and rents of every category. **Article 2, 5 lists the economic income among the sensitive data**: in Peru this column weighs the same as a diagnosis. The minimum wage was 1,130 soles a month from January 2025.

HOW IT IS FOUND

COLUMN NAME PE_FIN_INGRESOS – Path

90% `sueldo(_?(b[aá]sico|mensual|bruto|neto))?` `salario(_?(b[aá]sico|mensual|bruto|neto))?` `remuneraci[oó]n(_?(b[aá]sica|mensual|computable))?` `ingresos?(_?(mensuales?|totales?|familiares?|declarados?|netos?|econ[oó]micos?))?` `haberes` `honorarios` `renta(_?(bruta|neta|mensual|anual|de_?(primera|segunda|tercera|cuarta|quinta)_?categor[ií]a))?` `ingreso_?(base|familiar|percapita)` `salary` `income` `wages?`

DATA TYPE PE_FIN_INGRESOS – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
2800	→	**3433**

12.2 Credit history

PE_FIN_HISTORIAL_CREDITICIO

COLUMN NAME · REGEX DECOMPOSE

WHAT IT IS

The risk bureau report, the score, days in arrears, the SBS classification of the debtor — normal, con problema potencial, deficiente, dudoso, pérdida — and legal collection. The Ley 27489 governs the bureaus.

HOW IT IS FOUND

COLUMN NAME PE_FIN_HISTORIAL_CREDITICIO — Path

90% `historial_?(crediticio|de_?cr[eé]dito)` `central_?(de_?)?riesgo` `infocorp` `sentinel` `equifax` `experian` `xchange` `score(_?(crediticio|interno))?` `puntaje_?(de_?)?cr[eé]dito` `clasificaci[oó]n_?(sbs|crediticia|del_?deudor)` `calificaci[oó]n_?(de_?)?(riesgo|cr[eé]dito|cartera)` `categor[ií]a_?(de_?)?riesgo` `d[ií]as_?(de_?)?(mora|atraso)` `estado_?(de_?)?(cr[eé]dito|cartera|obligaci[oó]n)` `morosidad` `moroso` `cobranza_?judicial` `credit_?score` `credit_?rating`

HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a status is replaced by another from the list.

`PE_HISTORIAL_CREDITICIO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ESTADO_CREDITO`.

`PE_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

`PE_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`PE_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

`PE_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

`PE_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

`PE_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

`PE_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

`PE_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

`PE_ESTADO_CREDITO` SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-estados-credito.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Reportado en la central de riesgo	→	**Pérdida**
745	→	**471**

12.3 Debt

PE_FIN_DEUDA

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Credit and portfolio balances, amounts in arrears, instalments, disbursed amounts and limits.

HOW IT IS FOUND

COLUMN NAME PE_FIN_DEUDA – Path

85% `saldo_?(deuda|cr[eé]dito|capital|obligaci[oó]n|cartera|en_?mora|adeudado)` `monto_?(de_?la_?)?(deuda|cuota|obligaci[oó]n|cr[eé]dito|pr[eé]stamo|mora|desembolsado)` `deuda(_?total)?` `l[ií]mite_?(de_?)?(cr[eé]dito|tarjeta)` `l[ií]nea_?(de_?)?cr[eé]dito` `cuota_?mensual` `loan_?(amount|balance)` `outstanding_?balance` `debt`

DATA TYPE PE_FIN_DEUDA – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
18400	→	**69493**

12.4 Assets

PE_FIN_PATRIMONIO

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Net worth, assets, valuations and self-assessments, account, CTS and fund balances, term deposits and the sworn declaration of assets.

HOW IT IS FOUND

COLUMN NAME PE_FIN_PATRIMONIO – Path

85% `patrimonio(_?(neto|l[ií]quido))?` `activos?_?(totales?)?` `valor_?(del_?)?(inmueble|predio|veh[ií]culo|bienes|aval[uú]o|comercial)` `autoval[uú]o` `aval[uú]o` `saldo_?(cuenta|ahorro|dep[oó]sito|inversi[oó]n|cts|fondo)` `dep[oó]sito_?a_?plazo` `inversiones` `declaraci[oó]n_?jurada_?(de_?)?(bienes|ingresos)` `net_?worth` `account_?balance`

DATA TYPE PE_FIN_PATRIMONIO – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
148000	→	814723

12.5 Pension and benefits

PE_FIN_PENSION

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

WHAT IT IS

The amount of the pension, subsidy or benefit, and child support.

HOW IT IS FOUND

> COLUMN NAME PE_FIN_PENSION – Path
>
> 90% `pensi[oó]n(_?(monto|mensual|valor))?` `monto_?(pensi[oó]n|jubilaci[oó]n|beneficio|subsidio)` `valor_?(pensi[oó]n|subsidio|beneficio)` `haber_?(pensionable|jubilatorio)` `pensi[oó]n_?(de_?)?alimentos` `cuota_?alimentaria` `pension_?amount`

> DATA TYPE PE_FIN_PENSION – Type
>
> Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
1050	→	8262

12.6 Social programmes

PE_FIN_PROGRAMA_SOCIAL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Juntos, Pensión 65, Contigo, Cuna Más, Qali Warma, Beca 18, the FISE gas voucher and other programmes, by name or as a flag. Being in one says the household is poor.

HOW IT IS FOUND

> COLUMN NAME PE_FIN_PROGRAMA_SOCIAL – Path
>
> 90% `programa_?social` `beneficiario_?(programa|subsidio|bono|pensi[oó]n_?65)` `juntos` `pensi[oó]n_?65` `cuna_?m[aá]s` `qali_?warma` `beca_?18` `contigo` `trabaja_?per[uú]` `fise` `techo_?propio` `bono(_?(familiar|universal|yanapay|demogr[aá]fico))?` `subsidio(_?(gas|energ[ií]a|transporte|tipo))?` `transferencia_?monetaria` `ayuda_?(social|estatal)` `social_?programme`

VALUE FORMAT PE_FIN_PROGRAMA_SOCIAL — Values

PATTERN	STRENGTH	NOTES
One of 19 known values, ignoring case — Juntos, Pensión 65, Contigo, Cuna Más, Qali Warma, Beca 18, …	80%	
One of 10 known values, ignoring case — s, n, sí, no, 1, 2, …	30%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a programme from the list.

`PE_PROGRAMA_SOCIAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_PROGRAMA_SOCIAL_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_PROGRAMA_SOCIAL_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `pe-programas-sociales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Bono Yanapay	→	**Cuna Más**

12.7 SISFOH classification

PE_FIN_CLASIFICACION_SOCIOECONOMICA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The socioeconomic classification of the Padrón General de Hogares — extremely poor, poor, not poor — that decides who gets a subsidy, and the socioeconomic level or quintile.

HOW IT IS FOUND

COLUMN NAME PE_FIN_CLASIFICACION_SOCIOECONOMICA — Path

90% `clasificaci[oó]n_?socioecon[oó]mica` `csc` `sisfoh` `padr[oó]n_?general_?(de_?)?hogares` `nivel_?socioecon[oó]mico` `nse` `condici[oó]n_?(de_?)?pobreza` `pobreza` `quintil(_?(de_?)?(ingreso|riqueza))?` `estrato_?socioecon[oó]mico`

VALUE FORMAT PE_FIN_CLASIFICACION_SOCIOECONOMICA — Values

PATTERN	STRENGTH	NOTES
One of 12 known values, ignoring case — Pobre extremo, Pobre, No pobre, Sin clasificación, pobre, pobre extremo, …	60%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a class from the list.

`PE_CLASIFICACION_SOCIOECONOMICA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_CLASIFICACION_SOCIOECONOMICA_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CLASIFICACION_SOCIOECONOMICA_CODIGO` SECURE LOOKUP
>
> Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica-codigos.txt`, chosen by a keyed hash.
>
> `PE_CLASIFICACION_SOCIOECONOMICA_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Pobre extremo	→	**Sin clasificación**
2	→	**4**

12.8 Housing tenure

PE_FIN_VIVIENDA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Housing tenure in the categories the census uses: owned outright or being paid for, rented, lent by the employer or another household, owned by invasion.

HOW IT IS FOUND

COLUMN NAME PE_FIN_VIVIENDA — Path

90% `tenencia_?(de_?)?(la_?)?vivienda` `tipo_?(de_?)?tenencia` `vivienda_?(propia|alquilada|tipo_?tenencia)` `condici[oó]n_?(de_?)?(la_?)?vivienda` `ocupaci[oó]n_?vivienda` `housing_?tenure`

VALUE FORMAT PE_FIN_VIVIENDA — Values

PATTERN	STRENGTH	NOTES
One of 13 known values, ignoring case — Propia totalmente pagada, Propia pagándola a plazos, Alquilada, Cedida por el centro de trabajo u otro hogar, Propia por invasión, Otra, …	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced by another tenure from the list.

`PE_TENENCIA_VIVIENDA` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_TENENCIA_VIVIENDA_LISTA`.

> `PE_BANDERA` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_TENENCIA_VIVIENDA_LISTA` **SECURE LOOKUP**
>
> Replaces the value with one of the 6 values in `pe-tenencia-vivienda.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Ocupación de hecho	→	**Propia por invasión**

PART 13

Criminal records

The law does not list criminal records among the sensitive data, but the Constitution and the Código Procesal Penal treat them as reserved, and a record certificate in a test copy is a liability with no upside.

13.1 Criminal records

PE_PENAL_ANTECEDENTES

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

The criminal, police and judicial record certificate, the type of offence, the procedural situation, the requisitoria, being deprived of liberty, the INPE record and the REDAM register of maintenance debtors.

HOW IT IS FOUND

COLUMN NAME — PE_PENAL_ANTECEDENTES — Path

95% `antecedentes?(_?(penales|policiales|judiciales))?` `certificado_?(de_?)?antecedentes` `record_?policial` `reincidencia` `condena(_?(penal|tipo))?` `delito(_?(tipo|cometido))?` `tipo_?(de_?)?delito` `situaci[oó]n_?jur[ií]dica` `requisitoria` `orden_?(de_?)?captura` `privad[oa]_?(de_?)?libertad` `interno_?(penal|penitenciario)` `inpe` `reo` `deudor_?alimentario_?moroso` `redam` `criminal_?record`

55% `estado_?(del_?)?proceso` `sentencia` `fallo`

HOW IT IS MASKED

Replaced by a generic status from the list — no record, sentenced, acquitted —, whatever the outcome; a flag stays a flag.

`PE_ANTECEDENTES` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ANTECEDENTES_LISTA`.

> `PE_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `PE_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `PE_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `PE_ANTECEDENTES_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `pe-antecedentes.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Sentenciado por hurto en 2019	→	**Absuelto**

13.2 Court or prosecution file

PE_PENAL_EXPEDIENTE

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The number the Poder Judicial gives a case — correlative, year, judicial district and specialty, `00123-2024-0-1801-JR-CI-01` — and the carpeta fiscal of the Ministerio Público.

HOW IT IS FOUND

COLUMN NAME PE_PENAL_EXPEDIENTE – Path

90% `n(o|ro|um|umero)_?(de_?)?(expediente_?(judicial|penal|fiscal)|causa|proceso_?(judicial|penal)|denuncia|carpeta_?fiscal|referencia_?(judicial|fiscal))` `expediente_?(judicial|penal|fiscal)` `carpeta_?fiscal` `referencia_?(judicial|fiscal)` `causa_?penal` `n(o|ro|um|umero)_?[uú]nico_?(de_?)?expediente` `cui_?judicial`

50% `expediente` `causa` `proceso`

VALUE FORMAT PE_PENAL_EXPEDIENTE – Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{4,6}-\d{4}-\d-\d{4}-[A-Z\-]{2,12}-\d{2})`	90%	
`0*(?:\d{1,6}-[A-Z]{2,10}-\d{4})`	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The correlative and the court number are mapped digit by digit; year, judicial district and specialty stay. Other numbers are masked character by character.

`PE_EXPEDIENTE_JUDICIAL` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM` .

> `PE_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `PE_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
00123-2024-0-1801-JR-CI-01	→	**00957-2024-0-1801-JR-CI-06**
1234/2023	→	**9707/2023**

14

The DNI and RUC check digits

Both rules are modulus 11, and both end where Delphix's Check Digit algorithm cannot follow them on its own.

- **DNI**: weights 3, 2, 7, 6, 5, 4, 3 and 2 from the left; eleven minus the remainder (zero stays zero) gives a position from 0 to 10, which indexes two series — the digits 6 7 8 9 0 1 1 2 3 4 5 and the letters K A B C D E F G H I J. The letters belong to the documents issued up to 14 August 2007.
- **RUC**: weights 5, 4, 3, 2, 7, 6, 5, 4, 3 and 2 over the first ten digits; the digit is eleven minus the remainder, written 0 when that is ten and **1 when it is eleven**.

Check Digit writes *modulus minus remainder* and writes ten as `0`. Both rules agree with it everywhere except in the one place where that written zero is ambiguous: it stands for a remainder of 0 and for a remainder of 1 at once, and the two need different digits — 6 and 5 for the DNI, 1 and 0 for the RUC.

The policy resolves it with an **auxiliary Check Digit whose weights are the real ones doubled, modulo 11**. Doubling maps a remainder of 0 to 0 and a remainder of 1 to 2, so the auxiliary writes `0` for the first and `9` for the second — the two cases are told apart. A **Regex Decompose** then reads the pair of digits, writes the true one and drops the auxiliary. Four steps in a chain, no lookup table, and the whole body masked digit by digit before the digit is computed.

Because a natural person's RUC is the number 10 followed by the DNI, and both keep those eight digits under the same mapping, the masked DNI and the masked RUC of the same person still agree.

15

Where people live: distritos, UBIGEO and postal code

With the population the 2017 census counted and the location the INEI publishes, the policy knows the size, provincia, departamento and position of the 1,892 distritos and the 196 provincias.

- A **distrito** with fewer than 20,000 inhabitants becomes the nearest distrito of its **provincia** with at least 20,000; when its provincia has none, the nearest of its departamento. There are 1,633 of 1,892, with 7.4 million inhabitants.
- The **33 provincias** under 20,000 — Purús has 2,860 — follow the same rule inside their departamento. The 25 departamentos stay.
- Distance is measured between the census centroids, in a straight line. The eighteen distritos created after the census have no population or location of their own and take the centre of their provincia.
- Names repeat a great deal — San Juan, Santa Rosa, San José, Independencia, Progreso. A name is generalized only when every place bearing it is small, and then follows the most populous of them; when that bearer is a provincia, the replacement is a provincia. Written with the provincia or the departamento (`Balsas – Amazonas`, `Balsas (Amazonas)`) the same rule applies inside it.
- The **UBIGEO** follows the same table, so a person's distrito name and code still agree.
- The **centro poblado** and the **caserío** become `Zona rural`, the **urbanización** is replaced by another and the **postal code** keeps only the departamento.

Distrito names that are also given names or surnames, and departamento names, are left out of the detection list, so that a surname column is not taken for distritos.

16

Limits

- **Binary columns** — DNI photos, fingerprints, validation selfies, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **A verification character in a column of its own** is not recomputed: it is masked and no longer matches the masked DNI.
- **Departamento, the bank code and the phone area code stay.**
- **Data Cleansing leaves unknown values as they are**: a distrito spelled in a way that is not in the table — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces inside a sentence may not be found, and a name that is also a common word (`Rosa` , `Cruz` , `Flores`) is redacted wherever it appears.
- **Children and adolescents** — a birth date moves at most 60 days, so few people cross their 14th or 18th birthday, but some do. The regulation makes that line matter (art. 25): where it does, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, urbanización, distrito and postal code in separate columns are masked independently and no longer form a coherent address.
- **Masked bitcoin addresses** no longer pass their checksum: they serve to test the system, not to transact with it.
- **The engine key** links masked values back to production for whoever holds both: the copy is **dissociated** in the sense of article 41 of the regulation — a reversible procedure —, not anonymized, and remains personal data. Keep the key separate and safeguarded.

17

Verification

Before publication, the policy was checked outside Delphix:

- The DNI rule was checked against the worked example RENIEC's own routine is published with, and the RUC rule against RUCs published by public bodies.
- Every classifier configuration was reviewed against what Delphix accepts.
- Discovery was simulated on 145 columns with the whole policy at its assignment threshold — well named, badly named and 29 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes places was audited for all 1,892 distritos and 196 provincias: by name, by name with the provincia and the departamento, and by UBIGEO.
- The algorithms, run with the Delphix algorithm plugin, masked each data class's sample values and placeholders such as `N/A` , checking every output against its vocabulary; 150 DNIs of the digit series and 60 of the letter series, 200 RUCs of the four kinds of taxpayer for validity and collisions, and 20 DNI and RUC pairs to confirm they still agree. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the policy on real data, run a discovery job and a masking job on a representative copy.

18

Sources

- Ley 29733, Ley de Protección de Datos Personales (3 July 2011). Reglamento, Decreto Supremo 016-2024-JUS (El Peruano, 30 November 2024, in force 30 March 2025). Autoridad Nacional de Protección de Datos Personales, Ministerio de Justicia y Derechos Humanos.
- Ley 26842 General de Salud. Ley 30024 (Registro Nacional de Historias Clínicas Electrónicas). Ley 26626 (CONTRASIDA). Ley 29973 (persons with disabilities). Ley 27489 (risk bureaus). Ley 28592 (Registro Único de Víctimas). Ley 30364 (violence against women and the family group). Ley 29735 (indigenous languages). Ley 29785 (prior consultation).
- Instituto Nacional de Estadística e Informática: Censos Nacionales 2017 (XII de Población, VII de Vivienda y III de Comunidades Indígenas), Directorio Nacional de Centros Poblados, and the UBIGEO with the area and location of every distrito.
- RENIEC (DNI and the verification character). SUNAT (RUC and its check digit). SBS (debtor classification). MIDIS (SISFOH and the social programmes). Ministerio de Cultura (Base de Datos Oficial de Pueblos Indígenas u Originarios). JNE (Registro de Organizaciones Políticas). MTPE (minimum wage in force since January 2025). MTC (Placa Única Nacional de Rodaje). SERPOST (five-digit postal code).

APPENDIX A

Algorithms

Every algorithm of the policy in alphabetical order, with its configuration. The one a data class names is the top of its tree; the others are building blocks.

PE_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L2_ANIO_NACIMIENTO`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{3})(\d))`	keep · keep · `PE_CM_DIGITOS`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ANTECEDENTES_LISTA`.

Used by: `PE_PENAL_ANTECEDENTES`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_ANTECEDENTES_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `pe-antecedentes.txt`, chosen by a keyed hash.

Used by: `PE_ANTECEDENTES`

CONFIGURATION

lookupFile	`pe-antecedentes.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_APELLIDO NAME

Replaces the name with one of the 422 names in `pe-apellidos.txt`.

Used by: `PE_PALABRA_APELLIDO`

CONFIGURATION

lookupFile	pe-apellidos.txt
maskedValueCase	PRESERVE_INPUT
filterAccent	yes
inputCaseSensitive	no

PE_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PE_CM_ALFANUM .

Used by: PE_L1_APELLIDO , PE_NOMBRE_COMPLETO

PATTERNS, TRIED IN ORDER

(\S+)	PE_PALABRA_APELLIDO
(\S+)\s+(\S+)	PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO
(\S+)\s+(\S+)\s+(\S+)	PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO
(\S+)\s+(\S+)\s+(\S+)\s+(\S+)	PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO · PE_PALABRA_APELLIDO
otherwise	PE_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PE_APTITUD_LABORAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_APTITUD_LABORAL_LISTA .

Used by: PE_SALUD_OCUPACIONAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
(0*?)(?:(\d{1,6}))	keep · PE_CM_ALFANUM
otherwise	PE_APTITUD_LABORAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PE_APTITUD_LABORAL_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in pe-aptitud-laboral.txt , chosen by a keyed hash.

Used by: PE_APTITUD_LABORAL

CONFIGURATION

lookupFile	pe-aptitud-laboral.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_CATEGORIA_SUPRIMIDA`, `PE_ESTADO_CIVIL`, `PE_NIVEL_EDUCATIVO`, `PE_ORIGEN_ETNICO`, `PE_PUEBLO_INDIGENA`, `PE_LENGUA`, `PE_RELIGION`, `PE_IDEOLOGIA_POLITICA`, `PE_PARTIDO`, `PE_ORIENTACION_SEXUAL`, `PE_IDENTIDAD_GENERO`, `PE_COBERTURA_SALUD`, `PE_DIAGNOSTICO`, `PE_MEDICAMENTO`, `PE_DISCAPACIDAD`, `PE_APTITUD_LABORAL`, `PE_HISTORIAL_CREDITICIO`, `PE_PROGRAMA_SOCIAL`, `PE_CLASIFICACION_SOCIOECONOMICA`, `PE_TENENCIA_VIVIENDA`, `PE_ANTECEDENTES`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no)`	`PE_BANDERA_SI_NO`
`(?i)(true|false)`	`PE_BANDERA_TRUE_FALSE`
`(?i)(yes)`	`PE_BANDERA_YES_NO`
`(?i)([sn])`	`PE_BANDERA_S_N`
`(?i)([yx])`	`PE_BANDERA_Y_N`
`([01])`	`PE_BANDERA_0_1`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-0-1.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-0-1.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-s-n.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-s-n.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-si-no.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-si-no.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-true-false.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-true-false.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-y-n.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-y-n.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-bandera-yes-no.txt`, chosen by a keyed hash.

Used by: `PE_BANDERA`

CONFIGURATION

lookupFile	`pe-bandera-yes-no.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_SUPRIMIR`.

Used by: `PE_L3_CONVICCION_FILOSOFICA`, `PE_L3_AFILIACION_SINDICAL`, `PE_L3_VIDA_SEXUAL`, `PE_L3_VICTIMA`, `PE_SALUD_VIH`, `PE_SALUD_REPRODUCTIVA`, `PE_SALUD_MENTAL`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(\d{1,6})))))))))`	keep · keep · keep · keep · keep · keep · keep · `PE_CM_ALFANUM`
otherwise	`PE_SUPRIMIR`

CONFIGURATION

trimInput	yes
requireMask	no

PE_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in `pe-cie10.txt`, chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

CONFIGURATION

lookupFile	`pe-cie10.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `pe-cie10-decimal.txt`, chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

CONFIGURATION

lookupFile	`pe-cie10-decimal.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_CLASIFICACION_SOCIOECONOMICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_CLASIFICACION_SOCIOECONOMICA_LISTA`.

Used by: `PE_FIN_CLASIFICACION_SOCIOECONOMICA`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*)([1-9]\d{0,5}|0)`	keep · `PE_CLASIFICACION_SOCIOECONOMICA_CODIGO`
otherwise	`PE_CLASIFICACION_SOCIOECONOMICA_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_CLASIFICACION_SOCIOECONOMICA_CODIGO SECURE LOOKUP

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica-codigos.txt`, chosen by a keyed hash.

Used by: `PE_CLASIFICACION_SOCIOECONOMICA`

CONFIGURATION

lookupFile	`pe-clasificacion-socioeconomica-codigos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_CLASIFICACION_SOCIOECONOMICA_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in `pe-clasificacion-socioeconomica.txt`, chosen by a keyed hash.

Used by: `PE_CLASIFICACION_SOCIOECONOMICA`

CONFIGURATION

lookupFile	`pe-clasificacion-socioeconomica.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: `PE_L1_CARNE_EXTRANJERIA`, `PE_L1_PASAPORTE`, `PE_L1_ACTA`, `PE_L1_SEGURO_SOCIAL`, `PE_L1_CODIGO_ESTUDIANTE`, `PE_L1_COLEGIATURA`, `PE_L1_DIRECCION_COMPLEMENTO`, `PE_L1_BILLETERA`, `PE_L1_COOKIE`, `PE_L1_VEHICULO`, `PE_L1_INMUEBLE`, `PE_L1_CONTRATO`, `PE_L1_USUARIO`, `PE_SALUD_IDENTIFICADOR`, `PE_SALUD_RESULTADO_EXAMEN`, `PE_CATEGORIA_SUPRIMIDA`, `PE_DOCUMENTO`, `PE_NOMBRES`, `PE_APELLIDOS`, `PE_NOMBRE_COMPLETO`, `PE_EMAIL`, `PE_PUEBLO_INDIGENA`, `PE_LENGUA`, `PE_RELIGION`, `PE_IDEOLOGIA_POLITICA`, `PE_PARTIDO`, `PE_ORIENTACION_SEXUAL`, `PE_IDENTIDAD_GENERO`, `PE_MEDICAMENTO`, `PE_DISCAPACIDAD`, `PE_APTITUD_LABORAL`, `PE_PROGRAMA_SOCIAL`, `PE_TENENCIA_VIVIENDA`, `PE_ANTECEDENTES`, `PE_EXPEDIENTE_JUDICIAL`

CONFIGURATION

characterGroups	`0123456789` `AEIOU` `BCDFGHJKLMNPQRSTVWXYZ` `aeiou` `bcdfghjklmnpqrstvwxyz` `ÁÉÍÓÚÜ` `áéíóúü`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	yes

PE_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `PE_L1_CUENTA_BANCARIA`, `PE_RUC_CUERPO`, `PE_DOCUMENTO`, `PE_TELEFONO`, `PE_COORDENADA`, `PE_ANIO`, `PE_EDAD`, `PE_OCUPACION_O_CODIGO`, `PE_COBERTURA_SALUD`, `PE_PROCEDIMIENTO`, `PE_HISTORIAL_CREDITICIO`, `PE_EXPEDIENTE_JUDICIAL`

CONFIGURATION

characterGroups	`0123456789`
caseSensitive	yes

minMaskedPositions	`0`
preserveLeadingZeros	yes

PE_CM_DIGITOS_ANCHO_FIJO CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `PE_DNI_CD` , `PE_RUC_CUERPO`

CONFIGURATION

characterGroups	`0123456789`
caseSensitive	yes
minMaskedPositions	`0`

PE_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: `PE_L1_DIGITO_VERIFICACION` , `PE_L1_PLACA`

CONFIGURATION

characterGroups	`0123456789` `ABCDEFGHIJKLMNOPQRSTUVWXYZ`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	yes

PE_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: `PE_L1_DISPOSITIVO` , `PE_IP`

CONFIGURATION

characterGroups	`0123456789` `abcdef` `ABCDEF` `ghijklmnopqrstuvwxyz` `GHIJKLMNOPQRSTUVWXYZ`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	yes

PE_COBERTURA SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-cobertura-salud.txt` , chosen by a keyed hash.

Used by: `PE_COBERTURA_SALUD`

CONFIGURATION

lookupFile	`pe-cobertura-salud.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_COBERTURA_SALUD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_COBERTURA` .

Used by: `PE_SALUD_COBERTURA`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,8}))`	keep · `PE_CM_DIGITOS`
otherwise	`PE_COBERTURA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_CODIGO_POSTAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L2_CODIGO_POSTAL`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{2})(\d{3}))`	keep · keep · replace with `000`
`(0*?)(?:(\d{2})(\d{2}))`	keep · keep · replace with `00`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_CONTEO_LIMITADO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L2_PERSONAS_A_CARGO`

PATTERNS, TRIED IN ORDER

`(0*?)(?:([0-5]))`	keep · keep
`(0*?)(?:(\d+))`	keep · replace with `5`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L1_GEOLOCALIZACION`

PATTERNS, TRIED IN ORDER

`(-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+)`	keep · `PE_CM_DIGITOS` · keep · keep · `PE_CM_DIGITOS`
`(-?\d{1,3}[.,]\d)(\d+)`	keep · `PE_CM_DIGITOS`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PE_DIAGNOSTICO_TEXTO`.

Used by: `PE_SALUD_DIAGNOSTICO`

PATTERNS, TRIED IN ORDER

`([A-TV-Za-tv-z]\d{2}\.\d{1,2})`	`PE_CIE10_DECIMAL`
`([A-TV-Za-tv-z]\d{2,3}X?)`	`PE_CIE10`
`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
otherwise	`PE_DIAGNOSTICO_TEXTO`

CONFIGURATION

trimInput	yes
requireMask	no

PE_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in `pe-diagnosticos.txt`, chosen by a keyed hash.

Used by: `PE_DIAGNOSTICO`

CONFIGURATION

lookupFile	`pe-diagnosticos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_DIRECCION SECURE LOOKUP

Replaces the value with one of the 4,000 values in `pe-direcciones.txt`, chosen by a keyed hash.

Used by: `PE_L1_DIRECCION`

CONFIGURATION

lookupFile	`pe-direcciones.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_DISCAPACIDAD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_DISCAPACIDAD_LISTA`.

Used by: `PE_SALUD_DISCAPACIDAD`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_DISCAPACIDAD_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_DISCAPACIDAD_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-discapacidades.txt`, chosen by a keyed hash.

Used by: `PE_DISCAPACIDAD`

CONFIGURATION

lookupFile	`pe-discapacidades.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_DISTRITO DATA CLEANSING

Replaces each value found in `pe-distritos-generalizados.txt` (44,952 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PE_L2_DISTRITO`

CONFIGURATION

lookupFile	`pe-distritos-generalizados.txt`
delimiter	`|`
caseSensitive	yes
trimWhitespace	yes

PE_DNI_AUX CHECK DIGIT

Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6` from the right.

Used by: `PE_DNI_CON_DIGITO`, `PE_DNI_CON_LETRA`

CONFIGURATION

weightList	`0` `4` `6` `8` `10` `1` `3` `4` `6`
modulusNumber	`11`
checkDigitIndex	`9`
numDigitsForCheckdigitCalculation	`9`
calculateChecksumRightToLeft	yes
numericAlgorithm	`PE_IDENTIDAD`
alphaNumericAlgorithm	`PE_IDENTIDAD`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`
inputHandlingConfig.padCharacter	`0`
inputHandlingConfig.trimWhitespace	yes

PE_DNI_CD CHECK DIGIT

Masks the digits with `PE_CM_DIGITOS_ANCHO_FIJO` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 2 3` from the right.

Used by: `PE_DNI_CON_DIGITO`, `PE_DNI_CON_LETRA`

CONFIGURATION

weightList	`2 3 4 5 6 7 2 3`
modulusNumber	`11`
checkDigitIndex	`8`
numDigitsForCheckdigitCalculation	`8`
calculateChecksumRightToLeft	yes
numericAlgorithm	`PE_CM_DIGITOS_ANCHO_FIJO`
alphaNumericAlgorithm	`PE_CM_DIGITOS_ANCHO_FIJO`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`
inputHandlingConfig.padCharacter	`0`
inputHandlingConfig.trimWhitespace	yes

PE_DNI_CON_DIGITO STRING ALGORITHM CHAIN

Applies `PE_DNI_CD` → `PE_DNI_ESPACIO` → `PE_DNI_AUX` → `PE_DNI_VERIFICADOR`, in this order.

Used by: `PE_DOCUMENTO`

CONFIGURATION

algorithmReferences[1]	`PE_DNI_CD`
algorithmReferences[2]	`PE_DNI_ESPACIO`
algorithmReferences[3]	`PE_DNI_AUX`
algorithmReferences[4]	`PE_DNI_VERIFICADOR`

PE_DNI_CON_LETRA STRING ALGORITHM CHAIN

Applies `PE_DNI_CD` → `PE_DNI_ESPACIO` → `PE_DNI_AUX` → `PE_DNI_LETRA`, in this order.

Used by: `PE_DOCUMENTO`

CONFIGURATION

algorithmReferences[1]	`PE_DNI_CD`
algorithmReferences[2]	`PE_DNI_ESPACIO`
algorithmReferences[3]	`PE_DNI_AUX`
algorithmReferences[4]	`PE_DNI_LETRA`

PE_DNI_ESPACIO REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_DNI_CON_DIGITO`, `PE_DNI_CON_LETRA`

PATTERNS, TRIED IN ORDER

`(\d{8}[\s\-]?)(0)`	keep · replace with `00`
`(\d{8}[\s\-]?)(1)`	keep · replace with `10`
`(\d{8}[\s\-]?)(2)`	keep · replace with `20`

`(\d{8}[\s\-]?)(3)`	keep · replace with `30`
`(\d{8}[\s\-]?)(4)`	keep · replace with `40`
`(\d{8}[\s\-]?)(5)`	keep · replace with `50`
`(\d{8}[\s\-]?)(6)`	keep · replace with `60`
`(\d{8}[\s\-]?)(7)`	keep · replace with `70`
`(\d{8}[\s\-]?)(8)`	keep · replace with `80`
`(\d{8}[\s\-]?)(9)`	keep · replace with `90`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

PE_DNI_LETRA REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_DNI_CON_LETRA`

PATTERNS, TRIED IN ORDER

`(\d{8}[\s\-]?)(00)`	keep · replace with `K`
`(\d{8}[\s\-]?)(0\d)`	keep · replace with `J`
`(\d{8}[\s\-]?)(1\d)`	keep · replace with `A`
`(\d{8}[\s\-]?)(2\d)`	keep · replace with `B`
`(\d{8}[\s\-]?)(3\d)`	keep · replace with `C`
`(\d{8}[\s\-]?)(4\d)`	keep · replace with `D`
`(\d{8}[\s\-]?)(5\d)`	keep · replace with `E`
`(\d{8}[\s\-]?)(6\d)`	keep · replace with `F`
`(\d{8}[\s\-]?)(7\d)`	keep · replace with `G`
`(\d{8}[\s\-]?)(8\d)`	keep · replace with `H`
`(\d{8}[\s\-]?)(9\d)`	keep · replace with `I`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

PE_DNI_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_DNI_CON_DIGITO`

PATTERNS, TRIED IN ORDER

`(\d{8}[\s\-]?)(00)`	keep · replace with `6`
`(\d{8}[\s\-]?)(0\d)`	keep · replace with `5`
`(\d{8}[\s\-]?)(1\d)`	keep · replace with `7`
`(\d{8}[\s\-]?)(2\d)`	keep · replace with `8`
`(\d{8}[\s\-]?)(3\d)`	keep · replace with `9`
`(\d{8}[\s\-]?)(4\d)`	keep · replace with `0`
`(\d{8}[\s\-]?)(5\d)`	keep · replace with `1`
`(\d{8}[\s\-]?)(6\d)`	keep · replace with `1`

`(\d{8}[\s\-]?)(7\d)`	keep · replace with `2`
`(\d{8}[\s\-]?)(8\d)`	keep · replace with `3`
`(\d{8}[\s\-]?)(9\d)`	keep · replace with `4`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

Used by: `PE_L1_DNI`

PATTERNS, TRIED IN ORDER

`(0*?)(?:((?:10|15|17|20)\d{9}))`	keep · `PE_RUC`
`(0*?)(?:(\d{8}[\s\-]?[0-9]))`	keep · `PE_DNI_CON_DIGITO`
`(0*?)(?:(\d{8}[\s\-]?[A-JKa-jk]))`	keep · `PE_DNI_CON_LETRA`
`(0*?)(?:(\d{8}))`	keep · `PE_CM_DIGITOS`
otherwise	`PE_CM_ALFANUM`

CONFIGURATION

trimInput	yes
requireMask	no

PE_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L2_EDAD`

PATTERNS, TRIED IN ORDER

`(0*?)(?:([1-9]\d{2,}))`	keep · replace with `90`
`(0*?)(?:([1-9])(\d)([.,]\d+))`	keep · keep · `PE_CM_DIGITOS` · keep
`(0*?)(?:([1-9])(\d))`	keep · keep · `PE_CM_DIGITOS`
`(0*?)(?:(\d))`	keep · `PE_CM_DIGITOS`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

Used by: `PE_L1_EMAIL`

PATTERNS, TRIED IN ORDER

`([^@\s]+)@([^@\s]+)\.([A-Za-z]{2,24}(?:\.[A-Za-z]{2})?)`	`PE_CM_ALFANUM` · `PE_CM_ALFANUM` · replace with `test`
otherwise	`PE_CM_ALFANUM`

CONFIGURATION

trimInput	yes

requireMask	no

PE_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 24 values in `pe-empleadores.txt`, chosen by a keyed hash.

Used by: `PE_L2_EMPLEADOR`

CONFIGURATION

lookupFile	`pe-empleadores.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ESTADO_CIVIL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ESTADO_CIVIL_LISTA`.

Used by: `PE_L2_ESTADO_CIVIL`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*)([1-9]\d{0,5}|0)`	keep · `PE_ESTADO_CIVIL_CODIGO`
otherwise	`PE_ESTADO_CIVIL_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-estado-civil-codigos.txt`, chosen by a keyed hash.

Used by: `PE_ESTADO_CIVIL`

CONFIGURATION

lookupFile	`pe-estado-civil-codigos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-estado-civil.txt`, chosen by a keyed hash.

Used by: `PE_ESTADO_CIVIL`

CONFIGURATION

lookupFile	`pe-estado-civil.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ESTADO_CREDITO SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-estados-credito.txt`, chosen by a keyed hash.

Used by: `PE_HISTORIAL_CREDITICIO`

CONFIGURATION

lookupFile	`pe-estados-credito.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_EXPEDIENTE_JUDICIAL REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

Used by: `PE_PENAL_EXPEDIENTE`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{4,6})(-)(\d{4})(-)(\d)(-)(\d{4})(-)([A-Za-z\-]{2,12})(-)(\d{2}))`	keep · `PE_CM_DIGITOS` · keep · keep · keep · keep · keep · keep · keep · keep · keep · keep · `PE_CM_DIGITOS`
`(0*?)(?:(\d{1,6})([\s\-/])([A-Za-z]{2,10})([\s\-/])(\d{4}))`	keep · `PE_CM_DIGITOS` · keep · keep · keep · keep
`([A-Za-z]{1,6}[\s\-]?)(\d{1,6})([\s\-/])(\d{4})`	keep · `PE_CM_DIGITOS` · keep · keep
`(0*?)(?:(\d{1,6})([\s\-/])(\d{4}))`	keep · `PE_CM_DIGITOS` · keep · keep
otherwise	`PE_CM_ALFANUM`

CONFIGURATION

trimInput	yes
requireMask	no

PE_FECHA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `PE_L2_FECHA_EVENTO`

CONFIGURATION

minRange	`-30`
maxRange	`30`
unit	`DAYS`
roll	no

PE_FECHA_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: `PE_L2_FECHA_NACIMIENTO`

CONFIGURATION

minRange	`-60`
maxRange	`60`

unit	DAYS
roll	no

PE_HISTORIAL_CREDITICIO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_ESTADO_CREDITO.

Used by: PE_FIN_HISTORIAL_CREDITICIO

PATTERNS, TRIED IN ORDER

(0*?)(?:(\d{1,4}))	keep · PE_CM_DIGITOS
(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
otherwise	PE_ESTADO_CREDITO

CONFIGURATION

trimInput	yes
requireMask	no

PE_IDENTIDAD REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PE_DNI_AUX, PE_RUC_AUX

PATTERNS, TRIED IN ORDER

(.*)	keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_IDENTIDAD_GENERO_LISTA.

Used by: PE_L3_IDENTIDAD_GENERO

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
(0*?)(?:(\d{1,6}))	keep · PE_CM_ALFANUM
otherwise	PE_IDENTIDAD_GENERO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PE_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in pe-identidades-genero.txt, chosen by a keyed hash.

Used by: PE_IDENTIDAD_GENERO

CONFIGURATION

lookupFile	pe-identidades-genero.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_IDEOLOGIA_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_IDEOLOGIA_POLITICA_LISTA`.

Used by: `PE_L3_OPINION_POLITICA`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_IDEOLOGIA_POLITICA_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_IDEOLOGIA_POLITICA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `pe-ideologias-politicas.txt`, chosen by a keyed hash.

Used by: `PE_IDEOLOGIA_POLITICA`

CONFIGURATION

lookupFile	`pe-ideologias-politicas.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_INSTITUCION_EDUCATIVA SECURE LOOKUP

Replaces the value with one of the 12 values in `pe-instituciones-educativas.txt`, chosen by a keyed hash.

Used by: `PE_L2_INSTITUCION_EDUCATIVA`

CONFIGURATION

lookupFile	`pe-instituciones-educativas.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_CM_HEX`.

Used by: `PE_L1_IP`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{1,3})\.(\d{1,3})\.(\d{1,3})\.(\d{1,3}))`	keep · `PE_OCTETO` · `PE_OCTETO` · `PE_OCTETO` · `PE_OCTETO`

otherwise	`PE_CM_HEX`

CONFIGURATION

trimInput	yes
requireMask	no

PE_LENGUA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_LENGUA_LISTA`.

Used by: `PE_L3_LENGUA`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_LENGUA_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_LENGUA_LISTA SECURE LOOKUP

Replaces the value with one of the 18 values in `pe-lenguas.txt`, chosen by a keyed hash.

Used by: `PE_LENGUA`

CONFIGURATION

lookupFile	`pe-lenguas.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_MEDICAMENTO_LISTA`.

Used by: `PE_SALUD_MEDICAMENTO`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_MEDICAMENTO_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in `pe-medicamentos.txt`, chosen by a keyed hash.

Used by: `PE_MEDICAMENTO`

CONFIGURATION

lookupFile	`pe-medicamentos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_NACIONALIDAD SECURE LOOKUP

Replaces the value with one of the 14 values in `pe-nacionalidades.txt`, chosen by a keyed hash.

Used by: `PE_L3_NACIONALIDAD`

CONFIGURATION

lookupFile	`pe-nacionalidades.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_NIVEL_EDUCATIVO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_NIVEL_EDUCATIVO_LISTA`.

Used by: `PE_L2_NIVEL_EDUCATIVO`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*)([1-9]\d{0,5}|0)`	keep · `PE_NIVEL_EDUCATIVO_CODIGO`
otherwise	`PE_NIVEL_EDUCATIVO_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_NIVEL_EDUCATIVO_CODIGO SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo-codigos.txt`, chosen by a keyed hash.

Used by: `PE_NIVEL_EDUCATIVO`

CONFIGURATION

lookupFile	`pe-nivel-educativo-codigos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_NIVEL_EDUCATIVO_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-nivel-educativo.txt`, chosen by a keyed hash.

Used by: `PE_NIVEL_EDUCATIVO`

CONFIGURATION

lookupFile	`pe-nivel-educativo.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_NOMBRE NAME

Replaces the name with one of the 395 names in `pe-nombres.txt`.

Used by: `PE_PALABRA_NOMBRE`

CONFIGURATION

lookupFile	`pe-nombres.txt`
maskedValueCase	`PRESERVE_INPUT`
filterAccent	yes
inputCaseSensitive	no

PE_NOMBRE_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

Used by: `PE_L1_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

`([^,]+),\s*(.+)`	`PE_APELLIDOS` · `PE_NOMBRES`
`(\S+)`	`PE_PALABRA_NOMBRE`
`(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_APELLIDO`
`(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO`
`(\S+)\s+(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO`
`(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO`
`(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO` · `PE_PALABRA_APELLIDO`
otherwise	`PE_CM_ALFANUM`

CONFIGURATION

trimInput	yes
requireMask	no

PE_NOMBRES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PE_CM_ALFANUM`.

Used by: `PE_L1_NOMBRE`, `PE_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

`(\S+)`	`PE_PALABRA_NOMBRE`
`(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE`
`(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE`
`(\S+)\s+(\S+)\s+(\S+)\s+(\S+)`	`PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE` · `PE_PALABRA_NOMBRE`

otherwise	`PE_CM_ALFANUM`

CONFIGURATION

trimInput	yes
requireMask	no

PE_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: `PE_IP`

CONFIGURATION

minValue	`1`
maxValue	`254`

PE_OCUPACION SECURE LOOKUP

Replaces the value with one of the 39 values in `pe-ocupaciones.txt`, chosen by a keyed hash.

Used by: `PE_OCUPACION_O_CODIGO`

CONFIGURATION

lookupFile	`pe-ocupaciones.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_OCUPACION_O_CODIGO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_OCUPACION`.

Used by: `PE_L2_OCUPACION`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{2,5}))`	keep · `PE_CM_DIGITOS`
otherwise	`PE_OCUPACION`

CONFIGURATION

trimInput	yes
requireMask	no

PE_ORIENTACION_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ORIENTACION_SEXUAL_LISTA`.

Used by: `PE_L3_ORIENTACION_SEXUAL`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_ORIENTACION_SEXUAL_LISTA`

CONFIGURATION

trimInput	yes

requireMask	no

PE_ORIENTACION_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `pe-orientaciones-sexuales.txt`, chosen by a keyed hash.

Used by: `PE_ORIENTACION_SEXUAL`

CONFIGURATION

lookupFile	`pe-orientaciones-sexuales.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ORIGEN_ETNICO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_ORIGEN_ETNICO_LISTA`.

Used by: `PE_L3_ORIGEN_ETNICO`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*)([1-9]\d{0,5}|0)`	keep · `PE_ORIGEN_ETNICO_CODIGO`
otherwise	`PE_ORIGEN_ETNICO_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_ORIGEN_ETNICO_CODIGO SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-origen-etnico-codigos.txt`, chosen by a keyed hash.

Used by: `PE_ORIGEN_ETNICO`

CONFIGURATION

lookupFile	`pe-origen-etnico-codigos.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ORIGEN_ETNICO_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `pe-origen-etnico.txt`, chosen by a keyed hash.

Used by: `PE_ORIGEN_ETNICO`

CONFIGURATION

lookupFile	`pe-origen-etnico.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_PALABRA_APELLIDO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_APELLIDO`.

Used by: `PE_APELLIDOS`, `PE_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

`(?i)(de|del|la|las|los|y|e|san|santa|van|von|di|da)`	keep
otherwise	`PE_APELLIDO`

CONFIGURATION

trimInput	yes
requireMask	no

PE_PALABRA_NOMBRE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `PE_NOMBRE`.

Used by: `PE_NOMBRES`, `PE_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

`(?i)(de|del|la|las|los|y|e|san|santa|van|von|di|da)`	keep
otherwise	`PE_NOMBRE`

CONFIGURATION

trimInput	yes
requireMask	no

PE_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_PARTIDO_LISTA`.

Used by: `PE_L3_AFILIACION_PARTIDARIA`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_PARTIDO_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_PARTIDO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in `pe-partidos.txt`, chosen by a keyed hash.

Used by: `PE_PARTIDO`

CONFIGURATION

lookupFile	`pe-partidos.txt`
hashMethod	`SHA256`

maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_PROCEDIMIENTO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE_PROCEDIMIENTO_TEXTO.

Used by: PE_SALUD_PROCEDIMIENTO

PATTERNS, TRIED IN ORDER

(0*?)(?:(\d{4,6}))	keep · PE_CM_DIGITOS
otherwise	PE_PROCEDIMIENTO_TEXTO

CONFIGURATION

trimInput	yes
requireMask	no

PE_PROCEDIMIENTO_TEXTO SECURE LOOKUP

Replaces the value with one of the 10 values in pe-procedimientos.txt, chosen by a keyed hash.

Used by: PE_PROCEDIMIENTO

CONFIGURATION

lookupFile	pe-procedimientos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_PROGRAMA_SOCIAL_LISTA.

Used by: PE_FIN_PROGRAMA_SOCIAL

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
(0*?)(?:(\d{1,6}))	keep · PE_CM_ALFANUM
otherwise	PE_PROGRAMA_SOCIAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PE_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in pe-programas-sociales.txt, chosen by a keyed hash.

Used by: PE_PROGRAMA_SOCIAL

CONFIGURATION

lookupFile	pe-programas-sociales.txt

hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_PUEBLO_INDIGENA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_PUEBLO_INDIGENA_LISTA.

Used by: PE_L3_PUEBLO_INDIGENA

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
(0*?)(?:(\d{1,6}))	keep · PE_CM_ALFANUM
otherwise	PE_PUEBLO_INDIGENA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PE_PUEBLO_INDIGENA_LISTA SECURE LOOKUP

Replaces the value with one of the 21 values in pe-pueblos-indigenas.txt, chosen by a keyed hash.

Used by: PE_PUEBLO_INDIGENA

CONFIGURATION

lookupFile	pe-pueblos-indigenas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_RAZON_SOCIAL SECURE LOOKUP

Replaces the value with one of the 20 values in pe-razones-sociales.txt, chosen by a keyed hash.

Used by: PE_L1_RAZON_SOCIAL

CONFIGURATION

lookupFile	pe-razones-sociales.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PE_L1_CREDENCIAL, PE_L3_BIOMETRICO, PE_L3_GENETICO

PATTERNS, TRIED IN ORDER

`(.+)`	each character → `X`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PE_RELIGION_LISTA`.

Used by: `PE_L3_RELIGION`

PATTERNS, TRIED IN ORDER

`(?i)(s[ií]|no|true|false|yes|[snyx01])`	`PE_BANDERA`
`(0*?)(?:(\d{1,6}))`	keep · `PE_CM_ALFANUM`
otherwise	`PE_RELIGION_LISTA`

CONFIGURATION

trimInput	yes
requireMask	no

PE_RELIGION_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in `pe-religiones.txt`, chosen by a keyed hash.

Used by: `PE_RELIGION`

CONFIGURATION

lookupFile	`pe-religiones.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_RUC STRING ALGORITHM CHAIN

Applies `PE_RUC_CD` → `PE_RUC_ESPACIO` → `PE_RUC_AUX` → `PE_RUC_VERIFICADOR`, in this order.

Used by: `PE_RUC_COLUMNA`, `PE_DOCUMENTO`

CONFIGURATION

algorithmReferences[1]	`PE_RUC_CD`
algorithmReferences[2]	`PE_RUC_ESPACIO`
algorithmReferences[3]	`PE_RUC_AUX`
algorithmReferences[4]	`PE_RUC_VERIFICADOR`

PE_RUC_AUX CHECK DIGIT

Masks the digits with `PE_IDENTIDAD` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 4 6 8 10` from the right.

Used by: `PE_RUC`

CONFIGURATION

weightList	`0 4 6 8 10 1 3 4 6 8 10`

modulusNumber	11
checkDigitIndex	11
numDigitsForCheckdigitCalculation	11
calculateChecksumRightToLeft	yes
numericAlgorithm	PE_IDENTIDAD
alphaNumericAlgorithm	PE_IDENTIDAD
preserveRegex	[.\-\s]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

PE_RUC_CD CHECK DIGIT

Masks the digits with PE_RUC_CUERPO and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 2 3 4 5 from the right.

Used by: PE_RUC

CONFIGURATION

weightList	2 3 4 5 6 7 2 3 4 5
modulusNumber	11
checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	yes
numericAlgorithm	PE_RUC_CUERPO
alphaNumericAlgorithm	PE_RUC_CUERPO
preserveRegex	[.\-\s]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

PE_RUC_COLUMNA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE_RUC .

Used by: PE_L1_RUC

PATTERNS, TRIED IN ORDER

(0*?)(?:((?:10|15|17|20)\d{9}))	keep · PE_RUC
otherwise	PE_RUC

CONFIGURATION

trimInput	yes
requireMask	no

PE_RUC_CUERPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PE_CM_DIGITOS .

Used by: PE_RUC_CD

PATTERNS, TRIED IN ORDER

`(\d{2})(\d{8})`	keep · `PE_CM_DIGITOS_ANCHO_FIJO`
otherwise	`PE_CM_DIGITOS`

CONFIGURATION

trimInput	yes
requireMask	no

PE_RUC_ESPACIO REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_RUC`

PATTERNS, TRIED IN ORDER

`(\d{10})(0)`	keep · replace with `00`
`(\d{10})(1)`	keep · replace with `10`
`(\d{10})(2)`	keep · replace with `20`
`(\d{10})(3)`	keep · replace with `30`
`(\d{10})(4)`	keep · replace with `40`
`(\d{10})(5)`	keep · replace with `50`
`(\d{10})(6)`	keep · replace with `60`
`(\d{10})(7)`	keep · replace with `70`
`(\d{10})(8)`	keep · replace with `80`
`(\d{10})(9)`	keep · replace with `90`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_RUC_VERIFICADOR REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PE_RUC`

PATTERNS, TRIED IN ORDER

`(\d{10})(00)`	keep · replace with `1`
`(\d{10})(0\d)`	keep · replace with `0`
`(\d{10})(1\d)`	keep · replace with `1`
`(\d{10})(2\d)`	keep · replace with `2`
`(\d{10})(3\d)`	keep · replace with `3`
`(\d{10})(4\d)`	keep · replace with `4`
`(\d{10})(5\d)`	keep · replace with `5`
`(\d{10})(6\d)`	keep · replace with `6`
`(\d{10})(7\d)`	keep · replace with `7`
`(\d{10})(8\d)`	keep · replace with `8`
`(\d{10})(9\d)`	keep · replace with `9`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_SEXO REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PE_L2_SEXO

PATTERNS, TRIED IN ORDER

(?i)(femenino|masculino)	PE_SEXO_FEMENINO_MASCULINO
(?i)(mujer|hombre)	PE_SEXO_MUJER_HOMBRE
(?i)(female|male)	PE_SEXO_FEMALE_MALE
(?i)([fm])	PE_SEXO_F_M
(?i)([hm])	PE_SEXO_H_M
(0*?)(?:([12]))	keep · PE_SEXO_1_2
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_SEXO_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in pe-sexo-1-2.txt, chosen by a keyed hash.

Used by: PE_SEXO

CONFIGURATION

lookupFile	pe-sexo-1-2.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SEXO_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in pe-sexo-f-m.txt, chosen by a keyed hash.

Used by: PE_SEXO

CONFIGURATION

lookupFile	pe-sexo-f-m.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SEXO_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in pe-sexo-female-male.txt, chosen by a keyed hash.

Used by: PE_SEXO

CONFIGURATION

lookupFile	`pe-sexo-female-male.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SEXO_FEMENINO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-femenino-masculino.txt`, chosen by a keyed hash.

Used by: `PE_SEXO`

CONFIGURATION

lookupFile	`pe-sexo-femenino-masculino.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SEXO_H_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-h-m.txt`, chosen by a keyed hash.

Used by: `PE_SEXO`

CONFIGURATION

lookupFile	`pe-sexo-h-m.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SEXO_MUJER_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `pe-sexo-mujer-hombre.txt`, chosen by a keyed hash.

Used by: `PE_SEXO`

CONFIGURATION

lookupFile	`pe-sexo-mujer-hombre.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-sin-informacion.txt`, chosen by a keyed hash.

Used by: `PE_CATEGORIA_SUPRIMIDA`

CONFIGURATION

lookupFile	pe-sin-informacion.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PE_L1_TARJETA

PATTERNS, TRIED IN ORDER

(0*?)(?:(\d[\d\s\-]{11,22}\d))	keep · dlpx-core:Credit Card
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_TELEFONO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PE_CM_DIGITOS .

Used by: PE_L1_TELEFONO

PATTERNS, TRIED IN ORDER

(0*?)(?:(\+?51[\s\-]?)?(9)(\d{2})([\s\-]?)(\d{3})([\s\-]?)(\d{3}))	keep · keep · keep · PE_CM_DIGITOS · keep · PE_CM_DIGITOS · keep · PE_CM_DIGITOS
(\(?0?1\)?[\s\-]?)(\d{3})([\s\-]?)(\d{4})	keep · PE_CM_DIGITOS · keep · PE_CM_DIGITOS
(\(?0?[2-8]\d\)?[\s\-]?)(\d{3})([\s\-]?)(\d{3})	keep · PE_CM_DIGITOS · keep · PE_CM_DIGITOS
otherwise	PE_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

PE_TENENCIA_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PE_TENENCIA_VIVIENDA_LISTA .

Used by: PE_FIN_VIVIENDA

PATTERNS, TRIED IN ORDER

(?i)(s[ií]|no|true|false|yes|[snyx01])	PE_BANDERA
(0*?)(?:(\d{1,6}))	keep · PE_CM_ALFANUM
otherwise	PE_TENENCIA_VIVIENDA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PE_TENENCIA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `pe-tenencia-vivienda.txt`, chosen by a keyed hash.

Used by: `PE_TENENCIA_VIVIENDA`

CONFIGURATION

lookupFile	`pe-tenencia-vivienda.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with `[DATO]`, and the 1,906 terms of `pe-texto-nombres.txt` with `[NOMBRE]`.

Used by: `PE_L1_TEXTO_LIBRE`, `PE_SALUD_TEXTO_CLINICO`

PATTERNS, TRIED IN ORDER

`(?<!\d)\d{8}-[0-9A-JKa-jk](?!\d)`
`(?<!\d)\d{8}(?!\d)`
`(?<!\d)(10|15|17|20)\d{9}(?!\d)`
`[\w.+\-]+@[\w\-]+(\.[\w\-]+)+`
`(?<![\d\-])(\+?51[\s\-]?)?9\d{2}[\s\-]?\d{3}[\s\-]?\d{3}(?![\d\-])`
`(?<!\d)\d{13,19}(?!\d)`
`(?<![A-Za-z])[A-Z][0-9A-Z][A-Z][\s\-]?\d{3}(?![A-Za-z\d])`
`(?<!\d)(\d{1,3}\.){3}\d{1,3}(?!\d)`

CONFIGURATION

regExRedactValue	`[DATO]`
lookupFile	`pe-texto-nombres.txt`
lookupFileRedactValue	`[NOMBRE]`
isDenyList	yes

PE_UBIGEO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PE_L2_UBIGEO`

PATTERNS, TRIED IN ORDER

`(0*)([1-9]\d*)`	keep · `PE_UBIGEO_TABLA`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PE_UBIGEO_TABLA DATA CLEANSING

Replaces each value found in `pe-ubigeo-generalizado.txt` (2,533 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PE_UBIGEO`

CONFIGURATION

lookupFile	`pe-ubigeo-generalizado.txt`
delimiter	`|`
caseSensitive	yes
trimWhitespace	yes

PE_URBANIZACION SECURE LOOKUP

Replaces the value with one of the 104 values in `pe-urbanizaciones.txt`, chosen by a keyed hash.

Used by: `PE_L2_URBANIZACION`

CONFIGURATION

lookupFile	`pe-urbanizaciones.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PE_ZONA_RURAL SECURE LOOKUP

Replaces the value with one of the 1 values in `pe-zona-rural.txt`, chosen by a keyed hash.

Used by: `PE_L2_CENTRO_POBLADO`

CONFIGURATION

lookupFile	`pe-zona-rural.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

APPENDIX B

Files

The lookup and detection files the policy reads. In Delphix each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

FILE	LINES	USED BY	FIRST LINES
`pe-antecedentes.txt`	7	`PE_ANTECEDENTES_LISTA`	No registra antecedentes penales · Registra antecedentes · Procesado · Sentenciado
`pe-apellidos.txt`	422	`PE_APELLIDO`	Quispe · Mamani · Flores · Huamán
`pe-aptitud-laboral.txt`	4	`PE_APTITUD_LABORAL_LISTA`	Apto · Apto con restricciones · No apto · Pendiente
`pe-bandera-0-1.txt`	2	`PE_BANDERA_0_1`	0 · 1
`pe-bandera-s-n.txt`	2	`PE_BANDERA_S_N`	S · N
`pe-bandera-si-no.txt`	2	`PE_BANDERA_SI_NO`	Sí · No
`pe-bandera-true-false.txt`	2	`PE_BANDERA_TRUE_FALSE`	true · false
`pe-bandera-y-n.txt`	2	`PE_BANDERA_Y_N`	Y · N
`pe-bandera-yes-no.txt`	2	`PE_BANDERA_YES_NO`	Yes · No
`pe-cie10-decimal.txt`	33	`PE_CIE10_DECIMAL`	E11.9 · E78.5 · J45.9 · J06.9
`pe-cie10.txt`	37	`PE_CIE10`	I10 · E11 · E78 · J45
`pe-clasificacion-socioeconomica-codigos.txt`	4	`PE_CLASIFICACION_SOCIOECONOMICA_CODIGO`	1 · 2 · 3 · 4
`pe-clasificacion-socioeconomica.txt`	4	`PE_CLASIFICACION_SOCIOECONOMICA_LISTA`	Pobre extremo · Pobre · No pobre · Sin clasificación
`pe-cobertura-salud.txt`	8	`PE_COBERTURA`	EsSalud · SIS (Seguro Integral de Salud) · EPS · Sanidad de las Fuerzas Armadas
`pe-detectar-apellidos.txt`	519	`PE_L1_APELLIDO – List` `PE_L1_NOMBRE_COMPLETO – List`	Quispe · Mamani · Flores · Huamán
`pe-detectar-nacionalidades.txt`	41	`PE_L3_NACIONALIDAD – List`	Peruana · Venezolana · Colombiana · Ecuatoriana

FILE	LINES	USED BY	FIRST LINES
pe-detectar-nombres.txt	456	PE_L1_NOMBRE – List PE_L1_NOMBRE_COMPLETO – List	José · Jose · María · Maria
pe-detectar-sexo-letra.txt	3	PE_L2_SEXO – List	f · m · h
pe-detectar-sexo.txt	6	PE_L2_SEXO – List	femenino · masculino · mujer · hombre
pe-diagnosticos.txt	30	PE_DIAGNOSTICO_TEXTO	Hipertensión arterial · Diabetes mellitus tipo 2 · Asma · Lumbalgia
pe-direcciones.txt	4,000	PE_DIRECCION	Alameda Brasil 918 Int. 592 · Pasaje Pachacútec 1787, Urb. Zárate · Alameda Benavides 598, Urb. San Isidro · AA.HH. Nueva Esperanza, Jr. Brasil 787
pe-discapacidades.txt	8	PE_DISCAPACIDAD_LISTA	Física o motora · Auditiva · Visual · Sordoceguera
pe-distritos-generalizados.txt	44,952	PE_DISTRITO	Asunción|Los Baños del Inca · Asunción - Chachapoyas|Chachapoyas - Chachapoyas · Asunción, Chachapoyas|Chachapoyas, Chachapoyas · Asunción (Chachapoyas)|Chachapoyas (Chachapoyas)
pe-empleadores.txt	24	PE_EMPLEADOR	Distribuidora Andina S.A.C. · Constructora Pacífico S.A. · Transportes El Inca S.A.C. · Supermercados del Sur S.A.
pe-estado-civil-codigos.txt	6	PE_ESTADO_CIVIL_CODIGO	1 · 2 · 3 · 4
pe-estado-civil.txt	6	PE_ESTADO_CIVIL_LISTA	Soltero/a · Casado/a · Conviviente · Separado/a
pe-estados-credito.txt	11	PE_ESTADO_CREDITO	Normal · Con problema potencial · Deficiente · Dudoso
pe-identidades-genero.txt	7	PE_IDENTIDAD_GENERO_LISTA	Mujer cisgénero · Hombre cisgénero · Mujer trans · Hombre trans
pe-ideologias-politicas.txt	7	PE_IDEOLOGIA_POLITICA_LISTA	Izquierda · Centroizquierda · Centro · Centroderecha
pe-instituciones-educativas.txt	12	PE_INSTITUCION_EDUCATIVA	I.E. 1234 Nuestra Señora del Carmen · I.E. José Carlos Mariátegui · I.E.P. San Agustín · Colegio Nacional Mariscal Cáceres
pe-lenguas.txt	18	PE_LENGUA_LISTA	Castellano · Quechua · Aimara · Asháninka
pe-medicamentos.txt	22	PE_MEDICAMENTO_LISTA	Paracetamol · Ibuprofeno · Diclofenaco · Metformina
pe-nacionalidades.txt	14	PE_NACIONALIDAD	Peruana · Venezolana · Colombiana · Ecuatoriana
pe-nivel-educativo-codigos.txt	11	PE_NIVEL_EDUCATIVO_CODIGO	1 · 2 · 3 · 4
pe-nivel-educativo.txt	11	PE_NIVEL_EDUCATIVO_LISTA	Sin nivel · Inicial · Primaria incompleta · Primaria completa
pe-nombres.txt	395	PE_NOMBRE	José · María · Juan · Carlos

FILE	LINES	USED BY	FIRST LINES
pe-ocupaciones.txt	39	PE_OCUPACION	Empleado administrativo · Vendedor · Cajero · Chofer
pe-orientaciones-sexuales.txt	8	PE_ORIENTACION_SEXUAL_LISTA	Heterosexual · Gay · Lesbiana · Bisexual
pe-origen-etnico-codigos.txt	10	PE_ORIGEN_ETNICO_CODIGO	1 · 2 · 3 · 4
pe-origen-etnico.txt	10	PE_ORIGEN_ETNICO_LISTA	Quechua · Aimara · Nativo o indígena de la Amazonía · Perteneciente a otro pueblo indígena u originario
pe-partidos.txt	13	PE_PARTIDO_LISTA	Fuerza Popular · Perú Libre · Acción Popular · Alianza para el Progreso
pe-procedimientos.txt	10	PE_PROCEDIMIENTO_TEXTO	Consulta médica general · Consulta de emergencia · Hemograma completo · Glucosa en sangre
pe-programas-sociales.txt	10	PE_PROGRAMA_SOCIAL_LISTA	Juntos · Pensión 65 · Contigo · Cuna Más
pe-pueblos-indigenas.txt	21	PE_PUEBLO_INDIGENA_LISTA	Quechuas · Aimara · Asháninka · Awajún
pe-razones-sociales.txt	20	PE_RAZON_SOCIAL	Inversiones Andinas S.A.C. · Comercial Los Incas E.I.R.L. · Transportes Altiplano S.A.C. · Distribuidora del Pacífico S.R.L.
pe-religiones.txt	11	PE_RELIGION_LISTA	Católica · Evangélica · Cristiana · Adventista
pe-sexo-1-2.txt	2	PE_SEXO_1_2	1 · 2
pe-sexo-f-m.txt	2	PE_SEXO_F_M	F · M
pe-sexo-female-male.txt	2	PE_SEXO_FEMALE_MALE	Female · Male
pe-sexo-femenino-masculino.txt	2	PE_SEXO_FEMENINO_MASCULINO	Femenino · Masculino
pe-sexo-h-m.txt	2	PE_SEXO_H_M	H · M
pe-sexo-mujer-hombre.txt	2	PE_SEXO_MUJER_HOMBRE	Mujer · Hombre
pe-sin-informacion.txt	1	PE_SUPRIMIR	Sin información
pe-tenencia-vivienda.txt	6	PE_TENENCIA_VIVIENDA_LISTA	Propia totalmente pagada · Propia pagándola a plazos · Alquilada · Cedida por el centro de trabajo u otro hogar
pe-texto-nombres.txt	1,906	PE_TEXTO_LIBRE	José · JOSÉ · Jose · JOSE
pe-ubigeo-generalizado.txt	2,533	PE_UBIGEO_TABLA	010102|010101 · 010103|010101 · 010104|010101 · 010105|010101
pe-urbanizaciones.txt	104	PE_URBANIZACION	Urb. San Borja Norte · Urb. Las Gardenias · Urb. Monterrico · Urb. Chacarilla del Estanque
pe-zona-rural.txt	1	PE_ZONA_RURAL	Zona rural

APPENDIX C

Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

E1 `producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `jiron` `pasaje` `urbanizacion` `asentamiento` `caserio` `anexo` `centro` `poblado` `barrio` `distrito` `provincia` `departamento` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `essalud` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `lengua` `idioma` `lugar` `sede` `local` `tienda` `almacen` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `fiador` `avalista` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `imputado` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner` `completo` `apellidos?` `usuario` `corto` `largo`

E2 `producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `jiron` `pasaje` `urbanizacion` `asentamiento` `caserio` `anexo` `centro` `poblado` `barrio` `distrito` `provincia` `departamento` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `essalud` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `lengua` `idioma` `lugar` `sede` `local` `tienda` `almacen` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `fiador` `avalista` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `imputado` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner`

E3	`producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `jiron` `pasaje` `urbanizacion` `asentamiento` `caserio` `anexo` `centro` `poblado` `barrio` `distrito` `provincia` `departamento` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `essalud` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `lengua` `idioma` `lugar` `sede` `local` `tienda` `almacen` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `fiador` `avalista` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `imputado` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner` `full` `complete` `last` `first`
E4	`ip` `mac` `email` `e_?mail` `correo` `electr[oó]nica` `web` `url` `tipo` `id` `general` `regional`
E5	`ip` `mac` `email` `e_?mail` `web` `url` `type` `id`
E6	`id` `cd` `cod` `codigo` `tipo` `fecha` `estado` `num` `nro`
E7	`id` `tipo` `codigo` `fecha` `estado` `cola` `log`
E8	`cod` `codigo` `id` `judicial` `electoral` `fiscal` `escolar` `de_?riego`

APPENDIX D

Discovery policy and classifiers

The discovery policy `PE – Ley 29733 – v3` runs these 146 classifiers, with an assignment threshold of 60%.

Ley 29733 - datos personales y sensibles

CLASSIFIER	KIND	DATA CLASS
PE_L1_DNI – Path	COLUMN NAME	PE_L1_DNI
PE_L1_DNI – Type	DATA TYPE	PE_L1_DNI
PE_L1_DNI – Regex	VALUE FORMAT	PE_L1_DNI
PE_L1_RUC – Path	COLUMN NAME	PE_L1_RUC
PE_L1_RUC – Regex	VALUE FORMAT	PE_L1_RUC
PE_L1_DIGITO_VERIFICACION – Path	COLUMN NAME	PE_L1_DIGITO_VERIFICACION
PE_L1_DIGITO_VERIFICACION – Type	DATA TYPE	PE_L1_DIGITO_VERIFICACION
PE_L1_CARNE_EXTRANJERIA – Path	COLUMN NAME	PE_L1_CARNE_EXTRANJERIA
PE_L1_PASAPORTE – Path	COLUMN NAME	PE_L1_PASAPORTE
PE_L1_PASAPORTE – Regex	VALUE FORMAT	PE_L1_PASAPORTE
PE_L1_ACTA – Path	COLUMN NAME	PE_L1_ACTA
PE_L1_SEGURO_SOCIAL – Path	COLUMN NAME	PE_L1_SEGURO_SOCIAL
PE_L1_CODIGO_ESTUDIANTE – Path	COLUMN NAME	PE_L1_CODIGO_ESTUDIANTE
PE_L1_COLEGIATURA – Path	COLUMN NAME	PE_L1_COLEGIATURA
PE_L1_NOMBRE – Path	COLUMN NAME	PE_L1_NOMBRE
PE_L1_NOMBRE – List	KNOWN VALUES	PE_L1_NOMBRE
PE_L1_APELLIDO – Path	COLUMN NAME	PE_L1_APELLIDO
PE_L1_APELLIDO – List	KNOWN VALUES	PE_L1_APELLIDO
PE_L1_NOMBRE_COMPLETO – Path	COLUMN NAME	PE_L1_NOMBRE_COMPLETO
PE_L1_NOMBRE_COMPLETO – List	KNOWN VALUES	PE_L1_NOMBRE_COMPLETO
PE_L1_RAZON_SOCIAL – Path	COLUMN NAME	PE_L1_RAZON_SOCIAL
PE_L1_EMAIL – Path	COLUMN NAME	PE_L1_EMAIL
PE_L1_EMAIL – Type	DATA TYPE	PE_L1_EMAIL
PE_L1_EMAIL – Regex	VALUE FORMAT	PE_L1_EMAIL
PE_L1_TELEFONO – Path	COLUMN NAME	PE_L1_TELEFONO
PE_L1_TELEFONO – Regex	VALUE FORMAT	PE_L1_TELEFONO
PE_L1_DIRECCION – Path	COLUMN NAME	PE_L1_DIRECCION
PE_L1_DIRECCION – Type	DATA TYPE	PE_L1_DIRECCION
PE_L1_DIRECCION – Regex	VALUE FORMAT	PE_L1_DIRECCION
PE_L1_DIRECCION_COMPLEMENTO – Path	COLUMN NAME	PE_L1_DIRECCION_COMPLEMENTO
PE_L1_CUENTA_BANCARIA – Path	COLUMN NAME	PE_L1_CUENTA_BANCARIA
PE_L1_TARJETA – Path	COLUMN NAME	PE_L1_TARJETA

CLASSIFIER	KIND	DATA CLASS
PE_L1_TARJETA – Regex	VALUE FORMAT	PE_L1_TARJETA
PE_L1_BILLETERA – Path	COLUMN NAME	PE_L1_BILLETERA
PE_L1_BILLETERA – Regex	VALUE FORMAT	PE_L1_BILLETERA
PE_L1_IP – Path	COLUMN NAME	PE_L1_IP
PE_L1_IP – Regex	VALUE FORMAT	PE_L1_IP
PE_L1_DISPOSITIVO – Path	COLUMN NAME	PE_L1_DISPOSITIVO
PE_L1_DISPOSITIVO – Regex	VALUE FORMAT	PE_L1_DISPOSITIVO
PE_L1_COOKIE – Path	COLUMN NAME	PE_L1_COOKIE
PE_L1_PLACA – Path	COLUMN NAME	PE_L1_PLACA
PE_L1_PLACA – Regex	VALUE FORMAT	PE_L1_PLACA
PE_L1_VEHICULO – Path	COLUMN NAME	PE_L1_VEHICULO
PE_L1_VEHICULO – Regex	VALUE FORMAT	PE_L1_VEHICULO
PE_L1_INMUEBLE – Path	COLUMN NAME	PE_L1_INMUEBLE
PE_L1_CONTRATO – Path	COLUMN NAME	PE_L1_CONTRATO
PE_L1_USUARIO – Path	COLUMN NAME	PE_L1_USUARIO
PE_L1_CREDENCIAL – Path	COLUMN NAME	PE_L1_CREDENCIAL
PE_L1_CREDENCIAL – Regex	VALUE FORMAT	PE_L1_CREDENCIAL
PE_L1_GEOLOCALIZACION – Path	COLUMN NAME	PE_L1_GEOLOCALIZACION
PE_L1_GEOLOCALIZACION – Regex	VALUE FORMAT	PE_L1_GEOLOCALIZACION
PE_L1_TEXTO_LIBRE – Path	COLUMN NAME	PE_L1_TEXTO_LIBRE
PE_L1_TEXTO_LIBRE – Type	DATA TYPE	PE_L1_TEXTO_LIBRE
PE_L1_TEXTO_LIBRE – Regex	VALUE FORMAT	PE_L1_TEXTO_LIBRE
PE_L2_FECHA_NACIMIENTO – Path	COLUMN NAME	PE_L2_FECHA_NACIMIENTO
PE_L2_FECHA_NACIMIENTO – Type	DATA TYPE	PE_L2_FECHA_NACIMIENTO
PE_L2_ANIO_NACIMIENTO – Path	COLUMN NAME	PE_L2_ANIO_NACIMIENTO
PE_L2_ANIO_NACIMIENTO – Type	DATA TYPE	PE_L2_ANIO_NACIMIENTO
PE_L2_EDAD – Path	COLUMN NAME	PE_L2_EDAD
PE_L2_EDAD – Type	DATA TYPE	PE_L2_EDAD
PE_L2_FECHA_EVENTO – Path	COLUMN NAME	PE_L2_FECHA_EVENTO
PE_L2_FECHA_EVENTO – Type	DATA TYPE	PE_L2_FECHA_EVENTO
PE_L2_SEXO – Path	COLUMN NAME	PE_L2_SEXO
PE_L2_SEXO – List	KNOWN VALUES	PE_L2_SEXO
PE_L2_DISTRITO – Path	COLUMN NAME	PE_L2_DISTRITO
PE_L2_DISTRITO – Values	VALUE FORMAT	PE_L2_DISTRITO
PE_L2_UBIGEO – Path	COLUMN NAME	PE_L2_UBIGEO
PE_L2_UBIGEO – Regex	VALUE FORMAT	PE_L2_UBIGEO
PE_L2_CENTRO_POBLADO – Path	COLUMN NAME	PE_L2_CENTRO_POBLADO

CLASSIFIER	KIND	DATA CLASS
PE_L2_URBANIZACION – Path	COLUMN NAME	PE_L2_URBANIZACION
PE_L2_CODIGO_POSTAL – Path	COLUMN NAME	PE_L2_CODIGO_POSTAL
PE_L2_CODIGO_POSTAL – Type	DATA TYPE	PE_L2_CODIGO_POSTAL
PE_L2_ESTADO_CIVIL – Path	COLUMN NAME	PE_L2_ESTADO_CIVIL
PE_L2_ESTADO_CIVIL – Values	VALUE FORMAT	PE_L2_ESTADO_CIVIL
PE_L2_OCUPACION – Path	COLUMN NAME	PE_L2_OCUPACION
PE_L2_EMPLEADOR – Path	COLUMN NAME	PE_L2_EMPLEADOR
PE_L2_NIVEL_EDUCATIVO – Path	COLUMN NAME	PE_L2_NIVEL_EDUCATIVO
PE_L2_NIVEL_EDUCATIVO – Values	VALUE FORMAT	PE_L2_NIVEL_EDUCATIVO
PE_L2_INSTITUCION_EDUCATIVA – Path	COLUMN NAME	PE_L2_INSTITUCION_EDUCATIVA
PE_L2_PERSONAS_A_CARGO – Path	COLUMN NAME	PE_L2_PERSONAS_A_CARGO
PE_L2_PERSONAS_A_CARGO – Type	DATA TYPE	PE_L2_PERSONAS_A_CARGO
PE_L3_ORIGEN_ETNICO – Path	COLUMN NAME	PE_L3_ORIGEN_ETNICO
PE_L3_ORIGEN_ETNICO – Values	VALUE FORMAT	PE_L3_ORIGEN_ETNICO
PE_L3_PUEBLO_INDIGENA – Path	COLUMN NAME	PE_L3_PUEBLO_INDIGENA
PE_L3_PUEBLO_INDIGENA – Values	VALUE FORMAT	PE_L3_PUEBLO_INDIGENA
PE_L3_LENGUA – Path	COLUMN NAME	PE_L3_LENGUA
PE_L3_LENGUA – Values	VALUE FORMAT	PE_L3_LENGUA
PE_L3_NACIONALIDAD – Path	COLUMN NAME	PE_L3_NACIONALIDAD
PE_L3_NACIONALIDAD – List	KNOWN VALUES	PE_L3_NACIONALIDAD
PE_L3_RELIGION – Path	COLUMN NAME	PE_L3_RELIGION
PE_L3_RELIGION – Values	VALUE FORMAT	PE_L3_RELIGION
PE_L3_CONVICCION_FILOSOFICA – Path	COLUMN NAME	PE_L3_CONVICCION_FILOSOFICA
PE_L3_OPINION_POLITICA – Path	COLUMN NAME	PE_L3_OPINION_POLITICA
PE_L3_OPINION_POLITICA – Values	VALUE FORMAT	PE_L3_OPINION_POLITICA
PE_L3_AFILIACION_PARTIDARIA – Path	COLUMN NAME	PE_L3_AFILIACION_PARTIDARIA
PE_L3_AFILIACION_PARTIDARIA – Values	VALUE FORMAT	PE_L3_AFILIACION_PARTIDARIA
PE_L3_AFILIACION_SINDICAL – Path	COLUMN NAME	PE_L3_AFILIACION_SINDICAL
PE_L3_AFILIACION_SINDICAL – Values	VALUE FORMAT	PE_L3_AFILIACION_SINDICAL
PE_L3_BIOMETRICO – Path	COLUMN NAME	PE_L3_BIOMETRICO
PE_L3_BIOMETRICO – Type	DATA TYPE	PE_L3_BIOMETRICO
PE_L3_GENETICO – Path	COLUMN NAME	PE_L3_GENETICO
PE_L3_GENETICO – Regex	VALUE FORMAT	PE_L3_GENETICO
PE_L3_ORIENTACION_SEXUAL – Path	COLUMN NAME	PE_L3_ORIENTACION_SEXUAL
PE_L3_ORIENTACION_SEXUAL – Values	VALUE FORMAT	PE_L3_ORIENTACION_SEXUAL
PE_L3_VIDA_SEXUAL – Path	COLUMN NAME	PE_L3_VIDA_SEXUAL
PE_L3_IDENTIDAD_GENERO – Path	COLUMN NAME	PE_L3_IDENTIDAD_GENERO

CLASSIFIER	KIND	DATA CLASS
PE_L3_IDENTIDAD_GENERO – Values	VALUE FORMAT	PE_L3_IDENTIDAD_GENERO
PE_L3_VICTIMA – Path	COLUMN NAME	PE_L3_VICTIMA
PE_SALUD_IDENTIFICADOR – Path	COLUMN NAME	PE_SALUD_IDENTIFICADOR
PE_SALUD_COBERTURA – Path	COLUMN NAME	PE_SALUD_COBERTURA
PE_SALUD_COBERTURA – Values	VALUE FORMAT	PE_SALUD_COBERTURA
PE_SALUD_DIAGNOSTICO – Path	COLUMN NAME	PE_SALUD_DIAGNOSTICO
PE_SALUD_DIAGNOSTICO – Regex	VALUE FORMAT	PE_SALUD_DIAGNOSTICO
PE_SALUD_PROCEDIMIENTO – Path	COLUMN NAME	PE_SALUD_PROCEDIMIENTO
PE_SALUD_MEDICAMENTO – Path	COLUMN NAME	PE_SALUD_MEDICAMENTO
PE_SALUD_MEDICAMENTO – Values	VALUE FORMAT	PE_SALUD_MEDICAMENTO
PE_SALUD_TEXTO_CLINICO – Path	COLUMN NAME	PE_SALUD_TEXTO_CLINICO
PE_SALUD_TEXTO_CLINICO – Type	DATA TYPE	PE_SALUD_TEXTO_CLINICO
PE_SALUD_RESULTADO_EXAMEN – Path	COLUMN NAME	PE_SALUD_RESULTADO_EXAMEN
PE_SALUD_VIH – Path	COLUMN NAME	PE_SALUD_VIH
PE_SALUD_GRUPO_SANGUINEO – Path	COLUMN NAME	PE_SALUD_GRUPO_SANGUINEO
PE_SALUD_GRUPO_SANGUINEO – Values	VALUE FORMAT	PE_SALUD_GRUPO_SANGUINEO
PE_SALUD_DISCAPACIDAD – Path	COLUMN NAME	PE_SALUD_DISCAPACIDAD
PE_SALUD_DISCAPACIDAD – Values	VALUE FORMAT	PE_SALUD_DISCAPACIDAD
PE_SALUD_OCUPACIONAL – Path	COLUMN NAME	PE_SALUD_OCUPACIONAL
PE_SALUD_OCUPACIONAL – Values	VALUE FORMAT	PE_SALUD_OCUPACIONAL
PE_SALUD_REPRODUCTIVA – Path	COLUMN NAME	PE_SALUD_REPRODUCTIVA
PE_SALUD_MENTAL – Path	COLUMN NAME	PE_SALUD_MENTAL
PE_FIN_INGRESOS – Path	COLUMN NAME	PE_FIN_INGRESOS
PE_FIN_INGRESOS – Type	DATA TYPE	PE_FIN_INGRESOS
PE_FIN_HISTORIAL_CREDITICIO – Path	COLUMN NAME	PE_FIN_HISTORIAL_CREDITICIO
PE_FIN_DEUDA – Path	COLUMN NAME	PE_FIN_DEUDA
PE_FIN_DEUDA – Type	DATA TYPE	PE_FIN_DEUDA
PE_FIN_PATRIMONIO – Path	COLUMN NAME	PE_FIN_PATRIMONIO
PE_FIN_PATRIMONIO – Type	DATA TYPE	PE_FIN_PATRIMONIO
PE_FIN_PENSION – Path	COLUMN NAME	PE_FIN_PENSION
PE_FIN_PENSION – Type	DATA TYPE	PE_FIN_PENSION
PE_FIN_PROGRAMA_SOCIAL – Path	COLUMN NAME	PE_FIN_PROGRAMA_SOCIAL
PE_FIN_PROGRAMA_SOCIAL – Values	VALUE FORMAT	PE_FIN_PROGRAMA_SOCIAL
PE_FIN_CLASIFICACION_SOCIOECONOMICA – Path	COLUMN NAME	PE_FIN_CLASIFICACION_SOCIOECONOMICA
PE_FIN_CLASIFICACION_SOCIOECONOMICA – Values	VALUE FORMAT	PE_FIN_CLASIFICACION_SOCIOECONOMICA
PE_FIN_VIVIENDA – Path	COLUMN NAME	PE_FIN_VIVIENDA
PE_FIN_VIVIENDA – Values	VALUE FORMAT	PE_FIN_VIVIENDA

CLASSIFIER	KIND	DATA CLASS
`PE_PENAL_ANTECEDENTES – Path`	COLUMN NAME	`PE_PENAL_ANTECEDENTES`
`PE_PENAL_EXPEDIENTE – Path`	COLUMN NAME	`PE_PENAL_EXPEDIENTE`
`PE_PENAL_EXPEDIENTE – Regex`	VALUE FORMAT	`PE_PENAL_EXPEDIENTE`